

EDA Report

RIT 1410 AI Team 4A

Team Members:

ALLAM SRIDHAR (allamsridhar2001@gmail.com)
AHMED MAGDI (amg300571@gmail.com)
ALEX OJIBAH (Lex.ojibah@gmail.com)
RONEE HELEPI (roneehelep70@gmail.com)
VISHALINI S (svishalini04@gmail.com)
B SANDEEPKUMAR (battasandeep69@gmail.com)
RAVI TEJA BHUKYA (ravitejabhukya35@gmail.com)
QAIS ADENIJI-BELLO(qaismoyo19@gmail.com)
KIRSTEEN ARRIES (Kristen.arries99@gmail.com)
QAIS ADENJI BELLO (qaismoyo19@gmail.com)
MXOLISI MAHLANGU (Mxomahlangu3@gmail.com)

Introduction

Dataset Overview

The dataset consists of 8,558 records and 33 features, sourced from [the dashboard of Excelerate Platform]. Key features include:

- **Profile Id:** Unique identifier for each user.
- **Learner SignUp DateTime:** Timestamp for when the user signed up.
- **Opportunity Category:** Type of opportunity the user is engaging with like internship or course, etc..
- **Gender:** User's gender.
- **Current Student Status:** Indicates if the user is currently a student of Highschool or Undergraduate or Graduate or not in education.
- **Current/Intended Major:** The major the user is currently pursuing or intends to pursue.
- **Completion Date:** Date when the user completed the opportunity.
- **Opportunity Duration (days):** Duration of the opportunity in days.
- **Reward Amount :** The Scholarship amount given to the student who has completed his/her opportunity.
- **Skill Points Earned:** It specifies how many skill points were awarded to student on completion of opportunity.

This dataset offers insights into user engagement, completion trends, and demographic distributions.

Analysis Goals

The main objectives of this analysis are to:

- 1. To perform the EDA on the given Dataset and also to understand the Data, trends and patterns from the Data and present visualizations.
- 2. Examine trends in user signups and opportunity completions to assess engagement levels.
- 3. Investigate the relationship between demographic variables and completion rates to optimize user experiences.

Understanding these trends is vital for enhancing user engagement and success on the platform.

EDA

1. Univariate Analysis:

These are the results of Non Visual Univariate Analysis for each column:

Non-Visual Univariate Analysis:

Column: Profile Id

Data type: object
Number of unique values: 3964
Number of missing values: 0
Top 5 most common values:
Profile Id (no of occurances)
c5f56bba-7f74-4c24-8c23-67d91daa5e62 (16)
57f06936-a43f-47fc-8837-02b6ead734a5 (16)
c3b3ee41-b3d1-40b9-97d4-4011bb72e951 (16)
bd61f0a7-0d7a-4754-934d-17fcedf74392 (16)
93a89eda-ca80-4759-a753-a708859f52b9 (15)
Name: count, dtype: int64

Column: Learner SignUp DateTime

Data type: object
Number of unique values: 414
Number of missing values: 0
Top 5 most common values:
Learner SignUp DateTime (no of occurances)
2023-01-05 (146)
2023-07-09 (141)
2023-06-16 (118)
2024-01-08 (101)
2024-01-04 (98)
Name: count, dtype: int64

Column: Opportunity Id

Data type: object
Number of unique values: 23
Number of missing values: 0
Top 5 most common values:
Opportunity Id (no of occurances)
00000000-0GN2-A0AY-7XK8-C5FZPP (1423)

00000000-0GWQ-AXC5-X45C-2MHJ28 (980)
00000000-0GHB-4N83-QX9K-JM48K2 (836)
00000000-0GRA-BD28-CXVV-EYEX21 (784)
00000000-0GNT-FT74-MZT8-93VC0G (559)
Name: count, dtype: int64

Column: Opportunity Name

Data type: object
Number of unique values: 22
Number of missing values: 0
Top 5 most common values:
Opportunity Name (no of occurrences)
Career Essentials: Getting Started with Your Professional Journey (1423)
Data Visualization (980)
Project Management (836)
Health Care Management (784)
Digital Marketing (559)
Name: count, dtype: int64

Column: Opportunity Category

Data type: object
Number of unique values: 5
Number of missing values: 0
Top 5 most common values:
Opportunity Category (no of occurrences)
Internship (5421)
Course (2037)
Event (545)
Competition (425)
Engagement (130)
Name: count, dtype: int64

Column: Opportunity End Date

Data type: object
Number of unique values: 16
Number of missing values: 0
Top 5 most common values:
Opportunity End Date (no of occurrences)
2024-03-11 (4159)
2024-06-29 (1423)
2024-04-12 (784)
2024-04-11 (478)
2024-06-30 (389)
Name: count, dtype: int64

Column: Date of Birth

Data type: object
Number of unique values: 2620
Number of missing values: 0
Top 5 most common values:
Date of Birth (no of occurrences)
2001-05-20 (29)
2000-09-17 (28)
2002-09-17 (28)
2001-11-15 (25)
2000-06-21 (24)
Name: count, dtype: int64

Column: Gender

Data type: object

Number of unique values: 4

Number of missing values: 0

Top 5 most common values:

Gender (no of occurances)

Male (5018)

Female (3522)

Don't want to specify (15)

Other (3)

Name: count, dtype: int64

Column: Mobile Number

Data type: object

Number of unique values: 3834

Number of missing values: 0

Top 5 most common values:

Mobile Number (no of occurances)

+13145462111' (29)

+13144618335' (23)

+12164187367' (20)

+13144482010' (17)

+918374800652' (16)

Name: count, dtype: int64

Column: City

Data type: object

Number of unique values: 1156

Number of missing values: 0

Top 5 most common values:

City (no of occurances)

Saint Louis (3522)

Hyderabad (425)

Accra (110)

Guntur (102)

Bangalore (98)

Name: count, dtype: int64

Column: State

Data type: object

Number of unique values: 504

Number of missing values: 0

Top 5 most common values:

State (no of occurances)

Missouri (3313)

Telangana (738)

Andhra Pradesh (659)

MO (432)

Maharashtra (268)

Name: count, dtype: int64

Column: Country

Data type: object

Number of unique values: 71

Number of missing values: 0

Top 5 most common values:

Country (no of occurances)

United States (3969)
India (2843)
Nigeria (760)
Ghana (275)
Pakistan (219)
Name: count, dtype: int64

Column: Zip Code

Data type: object
Number of unique values: 1861
Number of missing values: 0
Top 5 most common values:
Zip Code (no of occurrences)
63108 (2278)
63146 (538)
63103 (414)
233 (223)
63043 (167)
Name: count, dtype: int64

Column: Institution Name

Data type: object
Number of unique values: 1764
Number of missing values: 0
Top 5 most common values:
Institution Name (no of occurrences)
Saint Louis University (4343)
Illinois Institute of Technology (138)
SRM Institute of Science and Technology (49)
Webster University (47)
Jawaharlal Nehru Technological University (33)
Name: count, dtype: int64

Column: Graduation Date

Data type: object
Number of unique values: 242
Number of missing values: 0
Top 5 most common values:
Graduation Date (no of occurrences)
2025-05-01 (2406)
2025-12-01 (657)
2024-12-01 (505)
2024-05-01 (319)
2025-08-01 (244)
Name: count, dtype: int64

Column: Current Student Status

Data type: object
Number of unique values: 4
Number of missing values: 0
Top 5 most common values:
Current Student Status (no of occurrences)
Graduate Program Student (5722)
Undergraduate Student (1799)
Not in Education (681)
High School Student (356)
Name: count, dtype: int64

Column: Current/Intended Major

Data type: object

Number of unique values: 406

Number of missing values: 0

Top 5 most common values:

Current/Intended Major (no of occurrences)

Information Systems (2163)

Computer Science (1052)

Health Data Science (371)

Cyber Security (326)

Computer Information Systems (250)

Name: count, dtype: int64

Column: Entry created at

Data type: object

Number of unique values: 1

Number of missing values: 0

Top 5 most common values:

Entry created at (no of occurrences)

2024-03-11 (8558)

Name: count, dtype: int64

Column: Status Description

Data type: object

Number of unique values: 8

Number of missing values: 0

Top 5 most common values:

Status Description

Rejected (3569)

Team Allocated (3276)

Started (767)

Dropped Out (617)

Waitlisted (109)

Name: count, dtype: int64

Column: Apply Date

Data type: object

Number of unique values: 442

Number of missing values: 0

Top 5 most common values:

Apply Date

2024-01-05 (160)

2024-01-08 (140)

2024-01-04 (126)

2024-01-24 (115)

2024-01-29 (110)

Name: count, dtype: int64

Column: Opportunity Start Date

Data type: object

Number of unique values: 26

Number of missing values: 0

Top 5 most common values:

Opportunity Start Date

2022-11-29 (3619)

2022-11-03 (1361)

2024-01-08 (988)

2022-12-30 (478)
2023-03-20 (389)
Name: count, dtype: int64

Column: Reward Awarded Date

Data type: object
Number of unique values: 9
Number of missing values: 0
Top 5 most common values:
Reward Awarded Date
1900-01-01 (8529)
2024-03-06 (11)
2024-02-12 (10)
2024-02-26 (2)
2024-02-14 (2)
Name: count, dtype: int64

Column: Completion Date

Data type: object
Number of unique values: 9
Number of missing values: 0
Top 5 most common values:
Completion Date
1900-01-01 (8529)
2024-03-06 (11)
2024-02-12 (10)
2024-02-26 (2)
2024-02-14 (2)
Name: count, dtype: int64

Column: Reward Amount

Data type: int64
Number of unique values: 3
Number of missing values: 0
Descriptive statistics:
count 8558.000000
mean 0.292124
std 6.250111
min 0.000000
25% 0.000000
50% 0.000000
75% 0.000000
max 200.000000
Name: Reward Amount, dtype: float64

Column: Badge Id

Data type: object
Number of unique values: 4
Number of missing values: 8529
Top 5 most common values:
Badge Id
00000000-10NF-MW89-S06E-JSHK5M (12)
00000000-10J8-464Q-NJ4M-H5GXV2 (8)
00000000-0GNH-7TCK-0QXP-EX3NY1 (7)
00000000-1030-NMJE-0HD7-W110RJ (2)
Name: count, dtype: int64

Column: Badge Name

Data type: object

Number of unique values: 4

Number of missing values: 8529

Top 5 most common values:

Badge Name

Startup Mastery Workshop Completed (12)

CPR Course Completion (8)

Career Essentials: Getting Started with Your Professional Journey (7)

Emotional Intelligence Course Completed (2)

Name: count, dtype: int64

Column: Skill Points Earned

Data type: int64

Number of unique values: 5

Number of missing values: 0

Descriptive statistics:

count 8558.000000

mean 1.230311

std 32.710013

min 0.000000

25% 0.000000

50% 0.000000

75% 0.000000

max 1052.000000

Name: Skill Points Earned, dtype: float64

Column: Skills Earned

Data type: object

Number of unique values: 4

Number of missing values: 8529

Top 5 most common values:

Skills Earned

["Critical Thinking","Collaboration","Leadership","Social Skills","Productivity","Initiative"] (12)

["Critical Thinking","Initiative","Flexibility","Leadership"] (8)

["Critical Thinking","Creative Thinking","Communication","Technology Literacy","Initiative"] (7)

["Critical Thinking","Communication","Flexibility","Leadership"] (2)

Name: count, dtype: int64

Column: Full_Address

Data type: object

Number of unique values: 3819

Number of missing values: 0

Top 5 most common values:

Full_Address

No address (43)

1158 Glenmeade Dr Apt B (29)

4404 McPherson Avenue (25)

3320 locust street #911 (24)

3952 Olive Street Apartment E (23)

Name: count, dtype: int64

Column: Full_Name

Data type: object

Number of unique values: 3844

Number of missing values: 0

Top 5 most common values:

Full_Name

Bhargavi Yarlagadda (19)
Sai Kowshik Kancharla (18)
Kalyan Varadagalla (17)
Harsha Vardhan Reddy Etikalapadu (16)
Varsha Kuchakulla (16)
Name: count, dtype: int64

Column: Age

Data type: int64
Number of unique values: 44
Number of missing values: 0
Descriptive statistics:
count 8558.000000
mean 24.588689
std 4.378603
min 5.000000
25% 22.000000
50% 24.000000
75% 26.000000
max 58.000000
Name: Age, dtype: float64

Column: Opportunity Duration (days)

Data type: int64
Number of unique values: 13
Number of missing values: 0
Descriptive statistics:
count 8558.000000
mean 409.936784
std 204.674331
min -7.000000
25% 468.000000
50% 468.000000
75% 468.000000
max 913.000000
Name: Opportunity Duration (days), dtype: float64

Column: Time to Apply (days)

Data type: int64
Number of unique values: 442
Number of missing values: 0
Descriptive statistics:
count 8558.000000
mean 107.421126
std 96.149610
min -30.000000
25% 36.000000
50% 65.000000
75% 181.000000
max 523.000000
Name: Time to Apply (days), dtype: float64

- Actually we replaced all the null values in the Reward Amount column and Skill Points Earned column with zero since we can understand that there were values in this columns only for those who had completed their opportunities and all the remaining null values implies that they haven't completed their opportunity yet.
- So if I don't consider those zeros and do non visual univariate analysis on those columns:

Skill Points Earned (Excluding 0):

count	29.000000
mean	363.068966
std	436.942292
min	50.000000
25%	50.000000
50%	137.000000
75%	1052.000000
max	1052.000000

Name: Skill Points Earned, dtype: float64

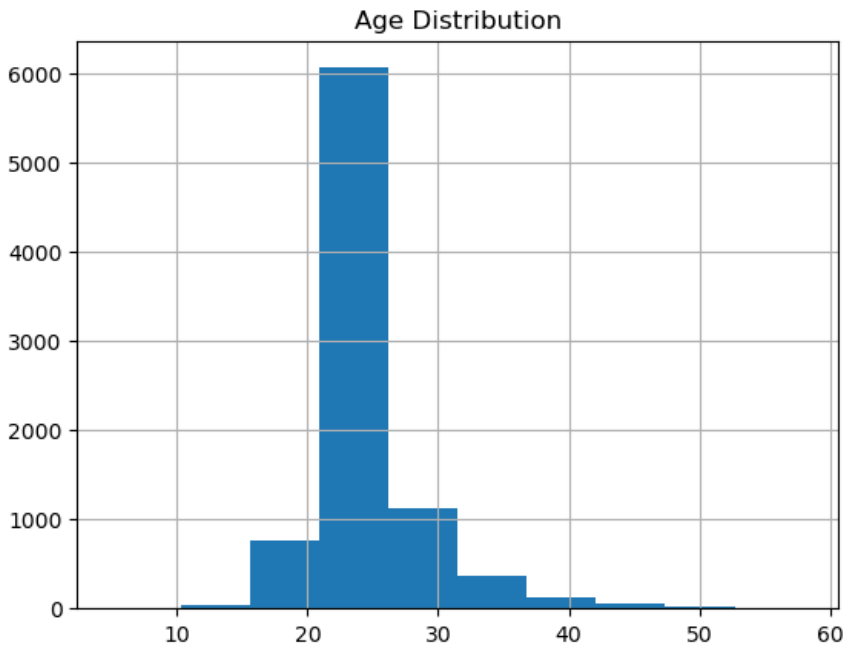
Reward Amount (Excluding 0):

count	29.000000
mean	86.206897
std	65.324126
min	50.000000
25%	50.000000
50%	50.000000
75%	50.000000
max	200.000000

Name: Reward Amount, dtype: float64

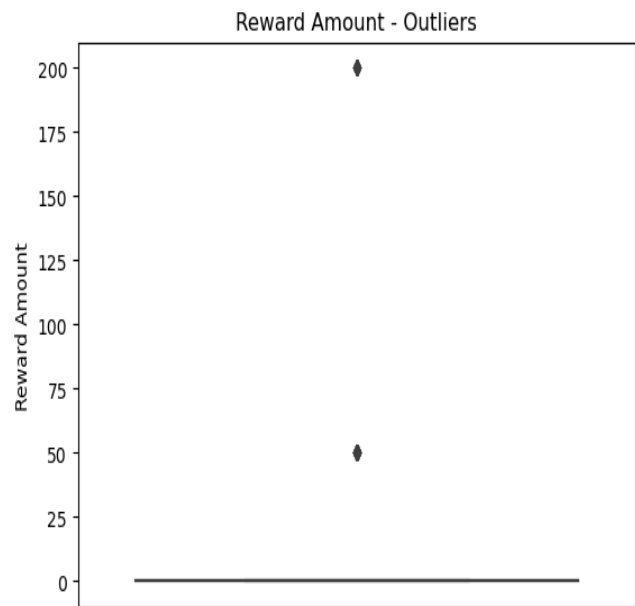
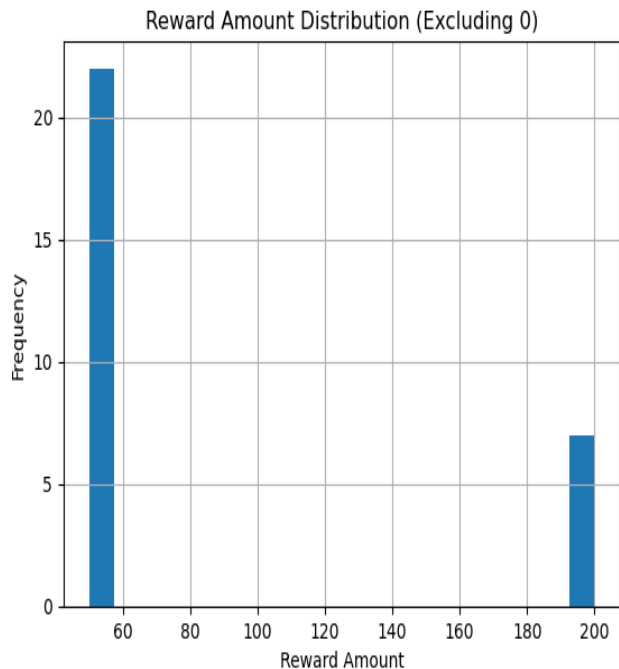
Visual Univariate Analysis:

Age Distribution:



- We can see that the distribution is slightly Right Skewed indicating that there are some outliers on the higher end. But most of the distribution is around 22-26 years.

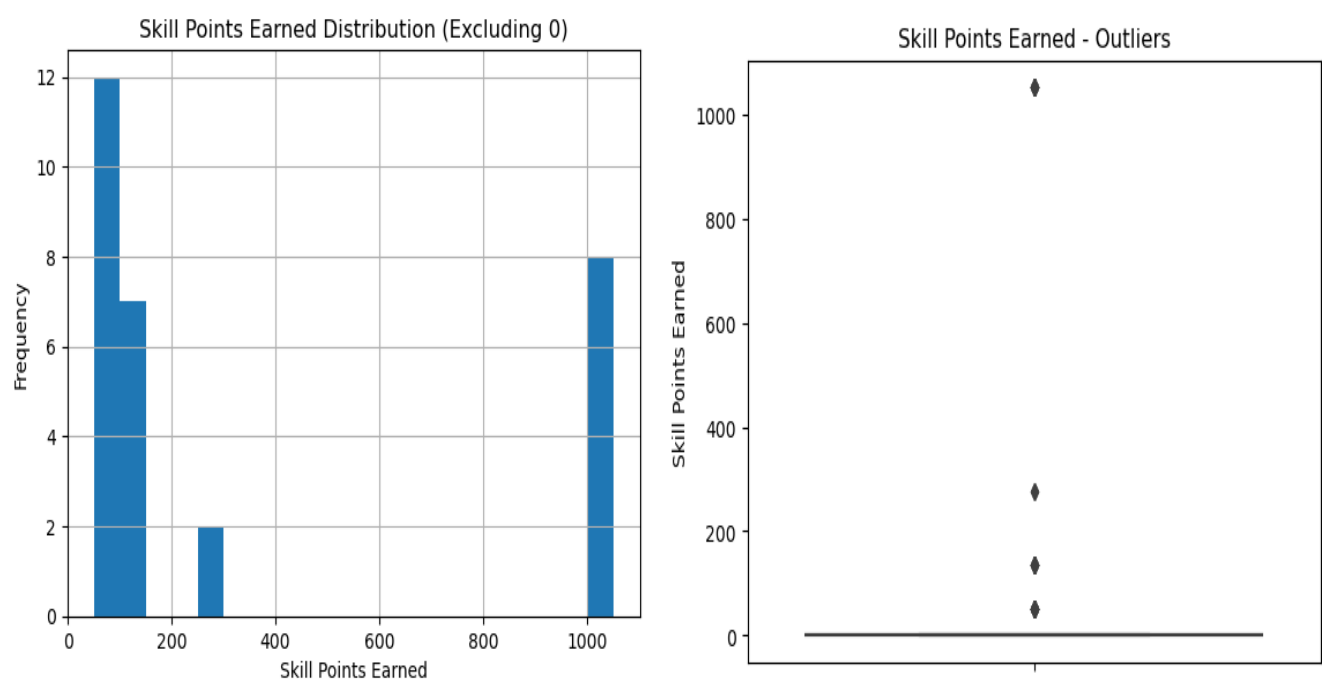
Reward Amount Distribution:



- We can clearly see the distribution is not continuous and there are values only at 50 and 200, indicating that the Reward Awarded was either 50 or 200 only.

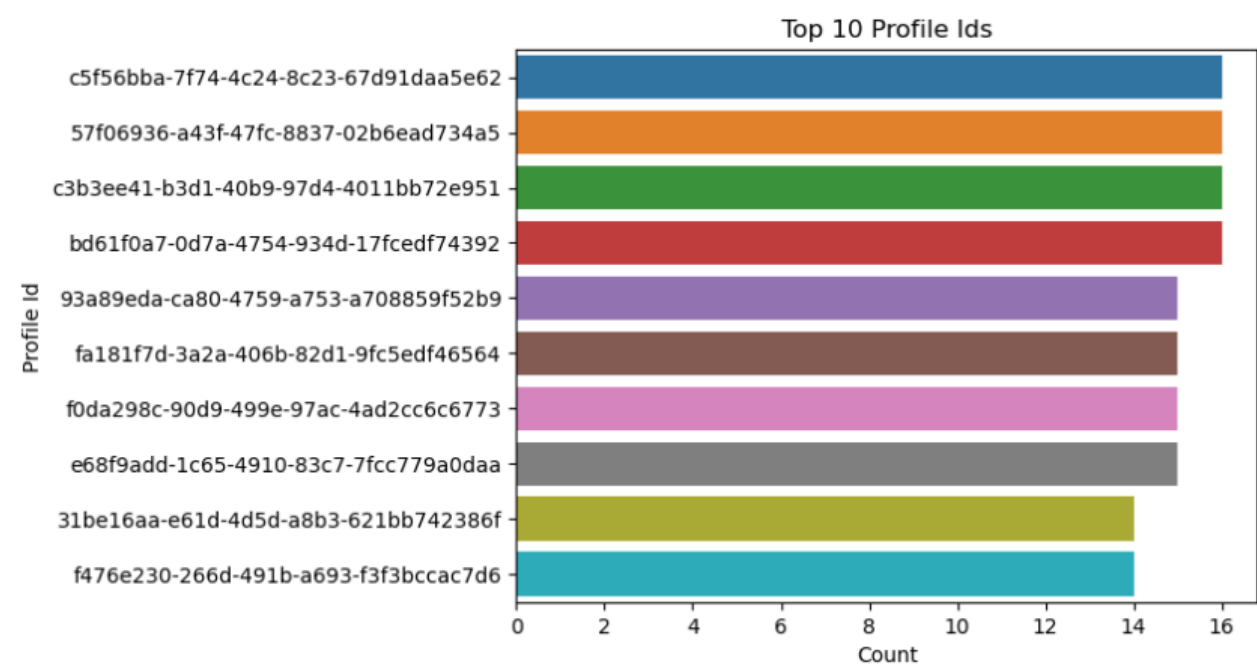
Skill Points Earned Distribution:

- We can clearly see that the distribution is not continuous and there are values only at 50,137,277 and 1052 only.



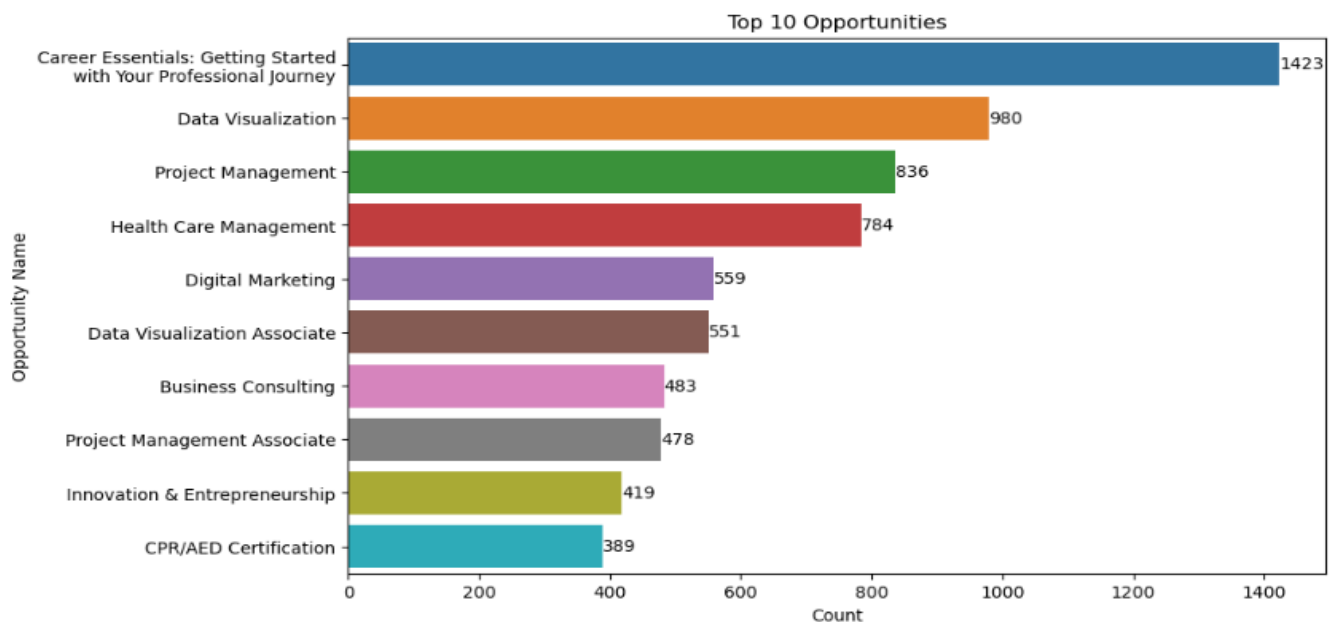
Profile Id Column:

- These are the Top 10 Profile Id's Which have been seen multiple times (but not duplicate), i.e these are the people who have applied for more than one opportunities.



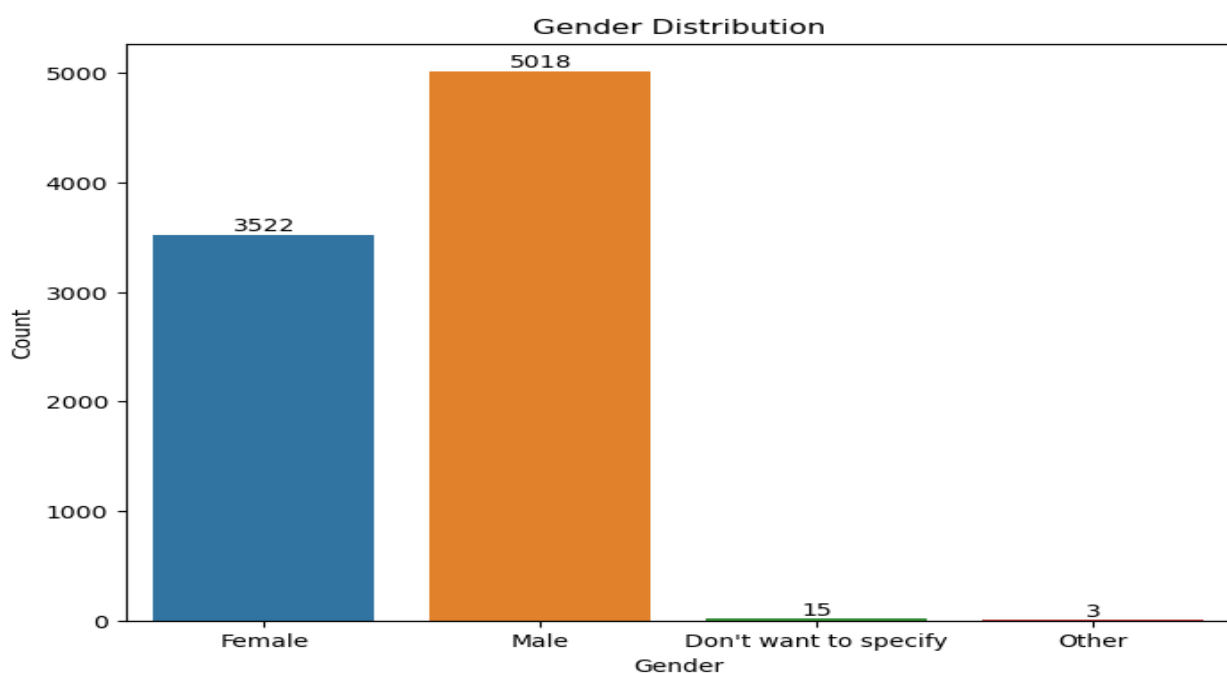
Opportunity Names Column:

- Here are the Top 10 Opportunities, to which most of the users have applied for:



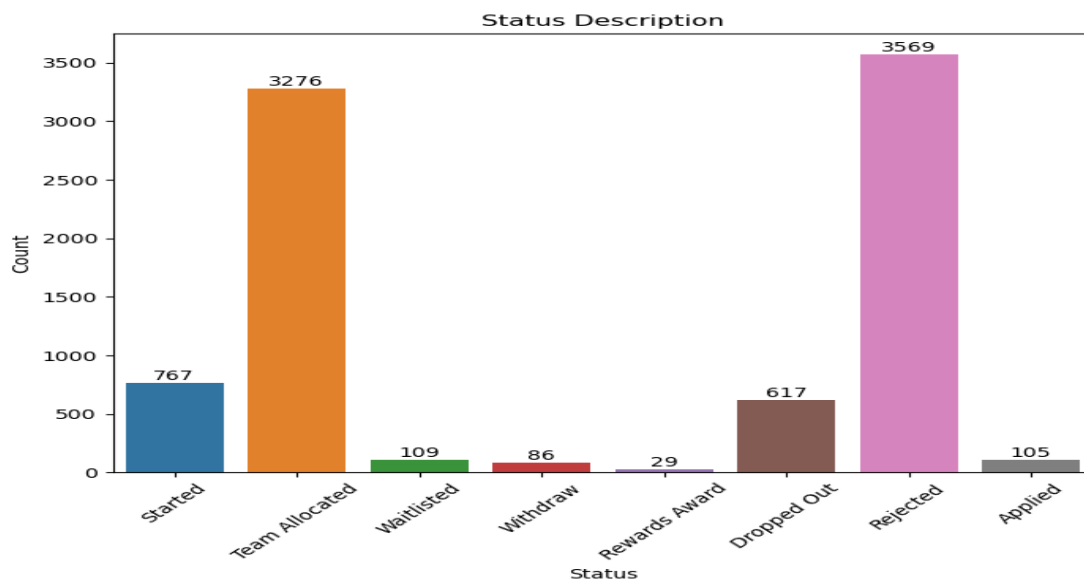
Gender Column:

- Here is the Gender Distribution on Opportunity Sign-ups.
- Here we can see that there is more male participation, compared to other genders



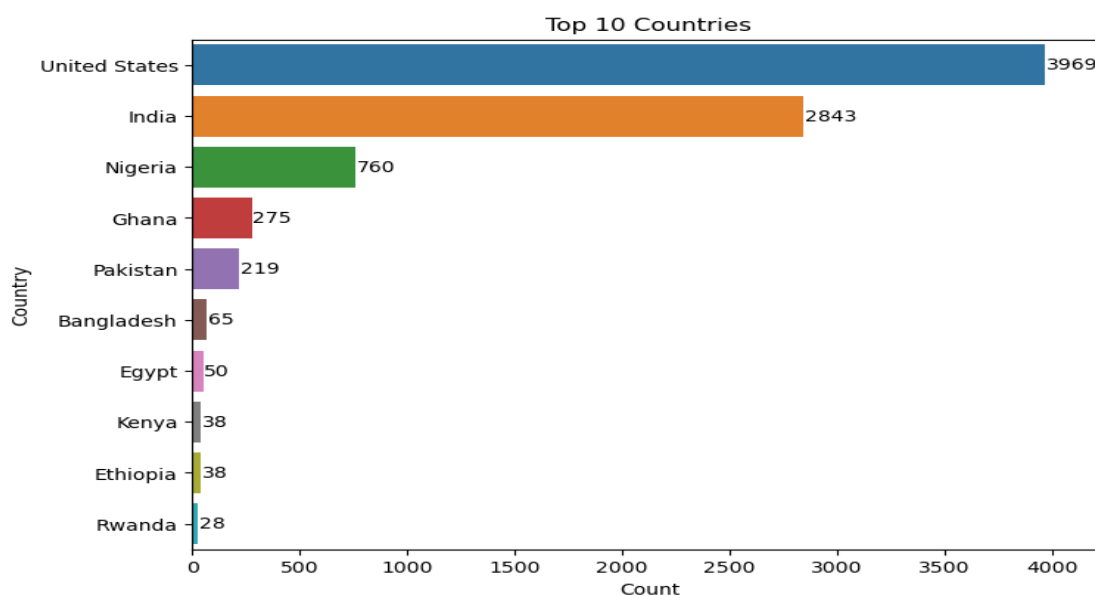
Status Distribution Column:

- Here we can see that Most of the Applications got rejected(3569)
- Apart from rejected ones from the remining most of them have been allocated (3276)
- For 767 of them the opportunity has been started, 617 have been dropped out and 86 have withdrawn
- For 105 people the status is still pending and for 109 people have been in Waiting List.



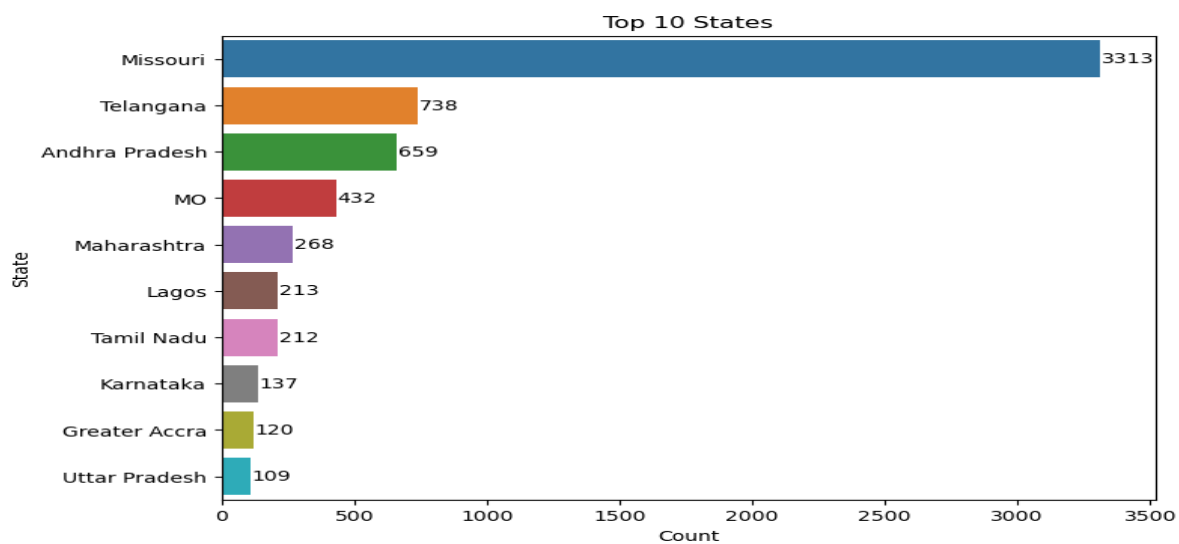
Country Column:

- Here is the distribution of Top 10 countries.



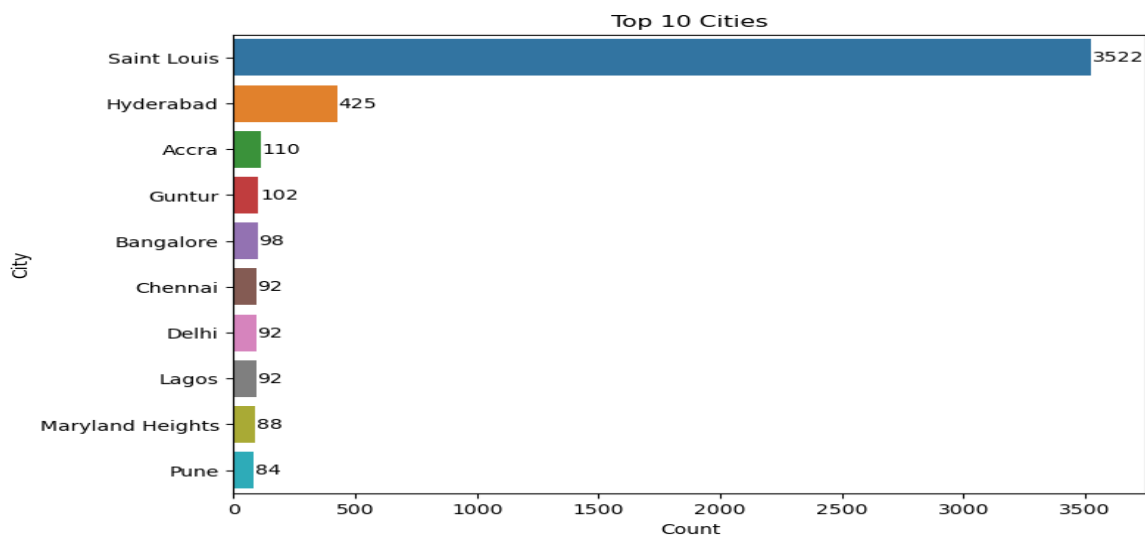
State Column:

- Here is the distribution of top 10 states.



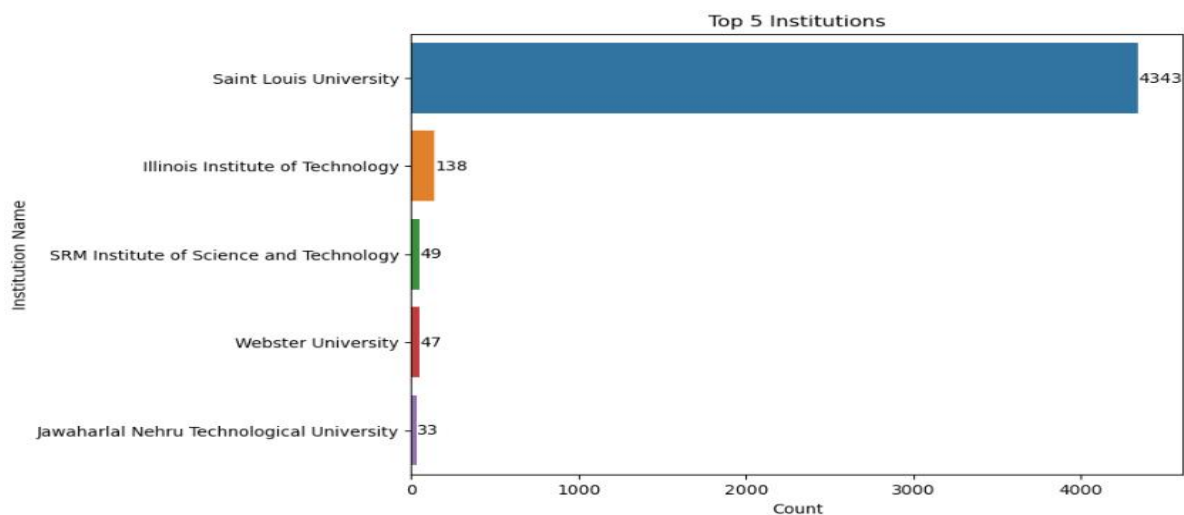
City Column:

- Here is the distribution of top 10 cities.



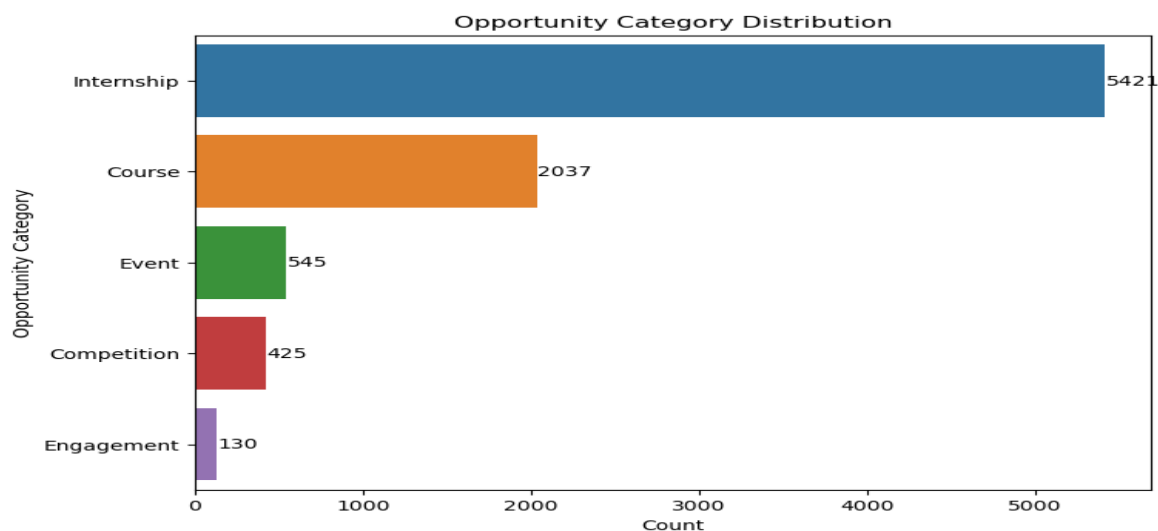
Institution Name Column:

- Here is the Distribution of Top 5 Institutions.



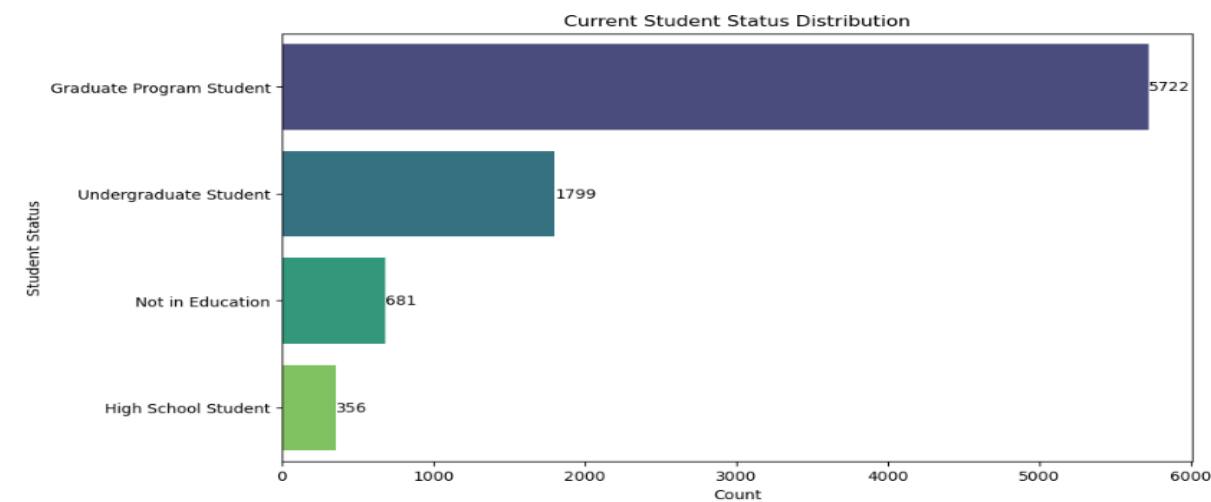
Opportunity Category Column:

- Here we can see that most of them have applied for Internships, few of them are interested in Courses and very few of them are interested in other categories.



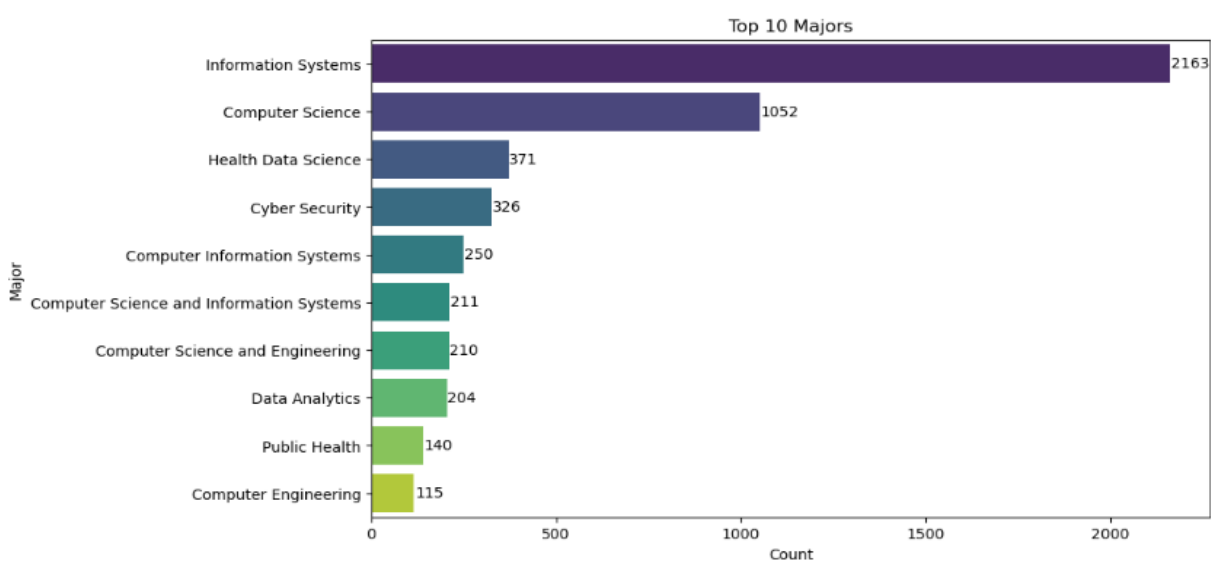
Current Student Status Column:

- Here we can see that there is huge participation from Graduate Program students.



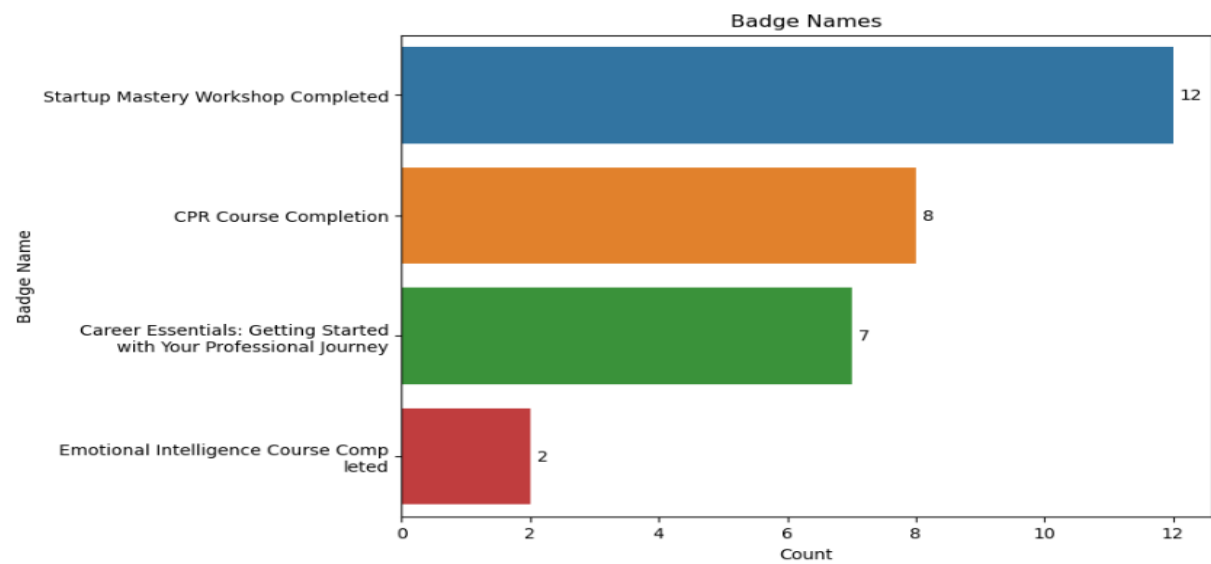
Current/Intended Majors Column:

- These are some of the top fields where the applicants are intrested in.



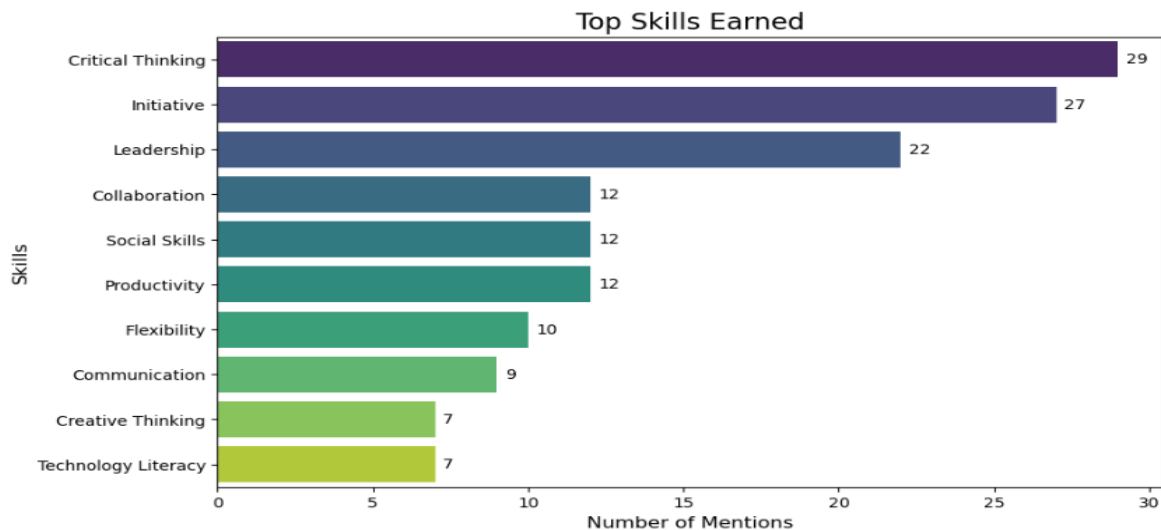
Badge Name Column:

- Here are the various badges that have been awarded to those who had completed their opportunity.



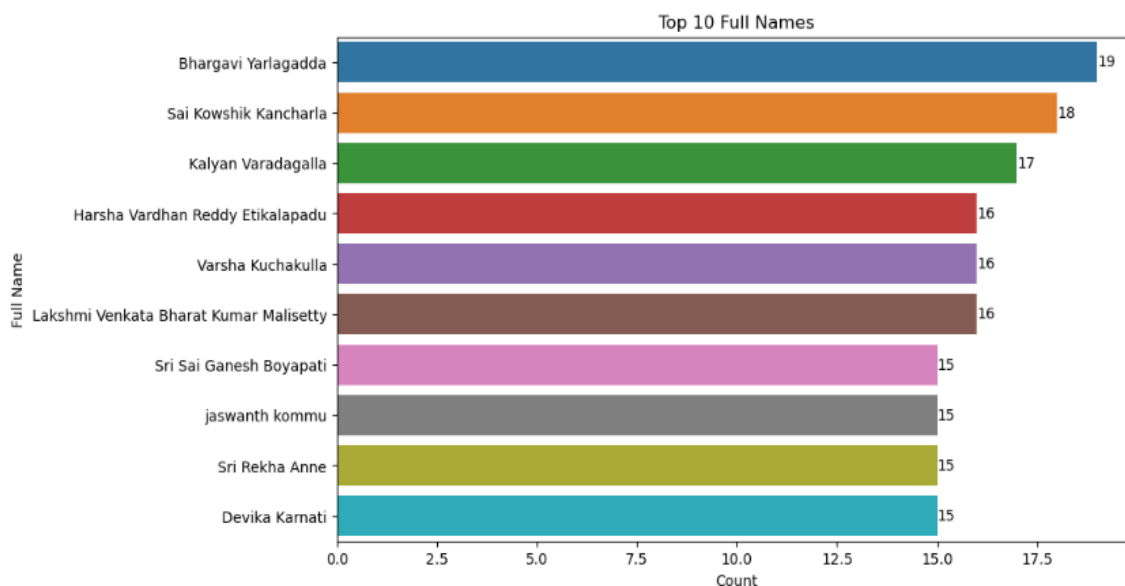
Skills Earned Column:

- Here are the top skills that the completed participants have achieved.



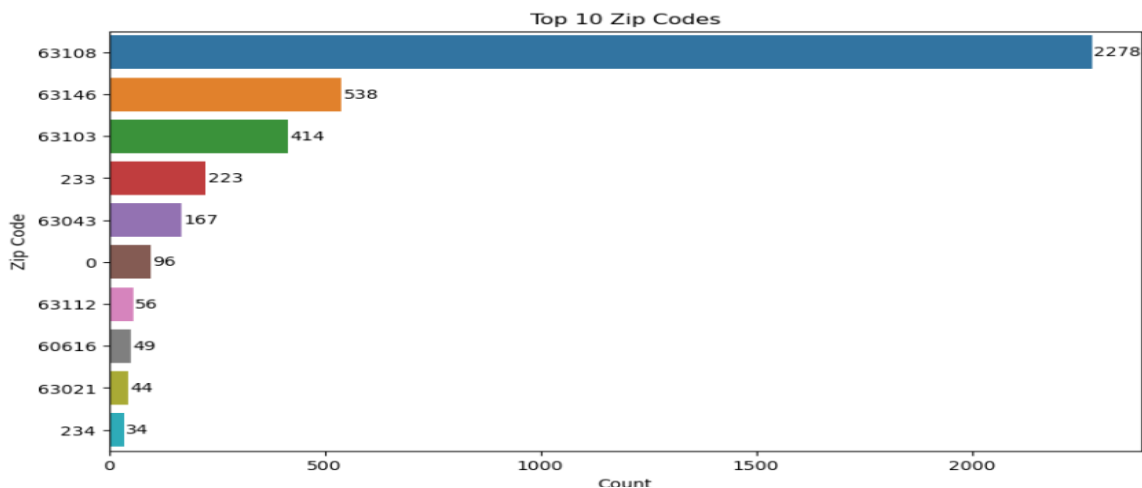
Full Name Column:

- These are the some of the people who have applied for most of the opportunities.



Zip Codes:

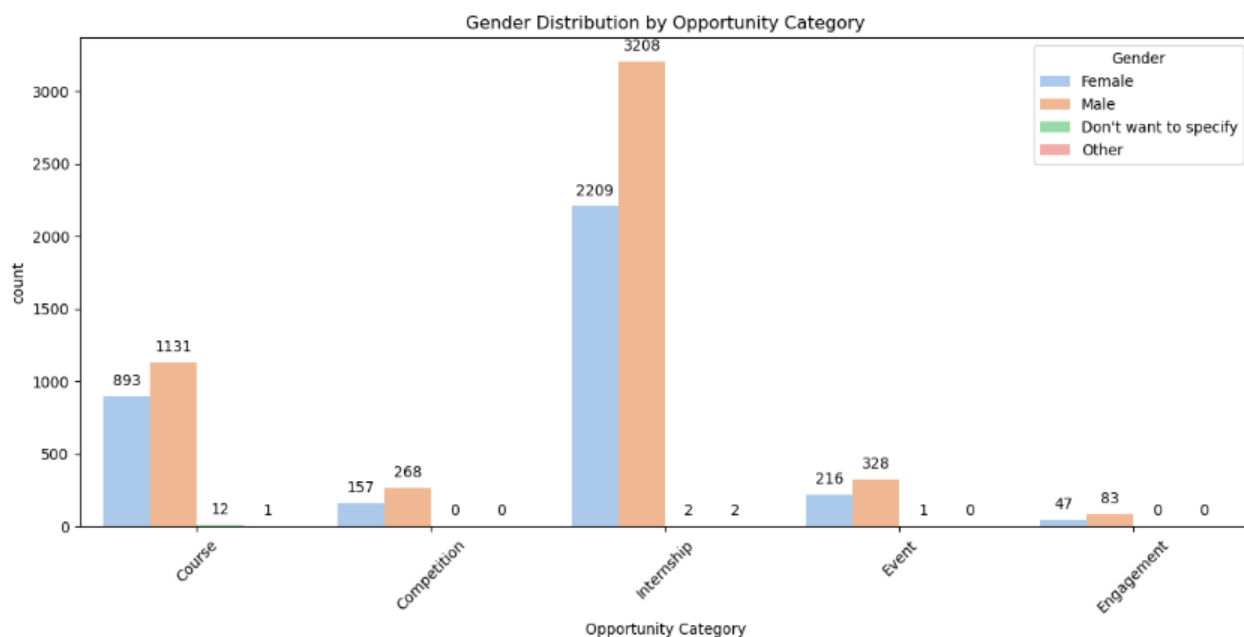
- These are top 10 zip codes



Bivariate Analysis:

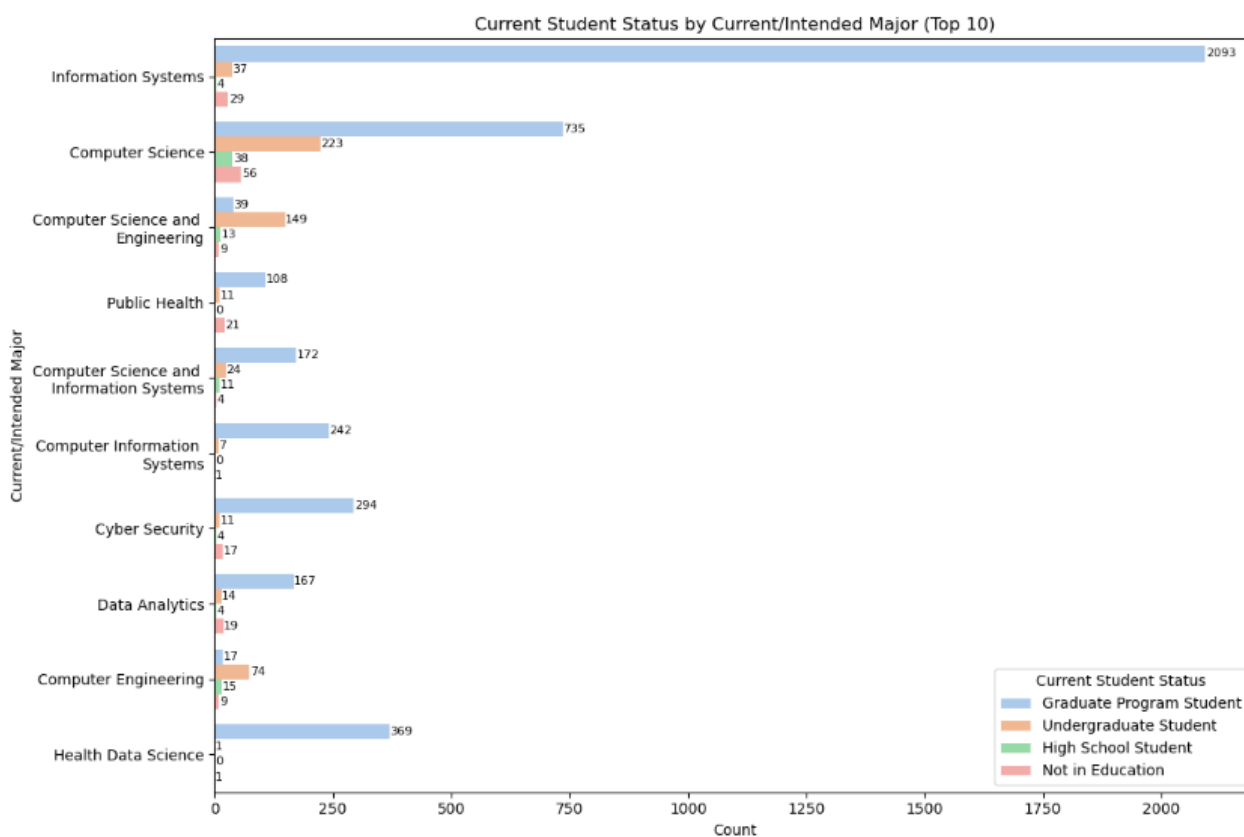
Gender vs Opportunity Category:

- Here we can see that in all the opportunity Categories there is more participation of Male than the other Genders



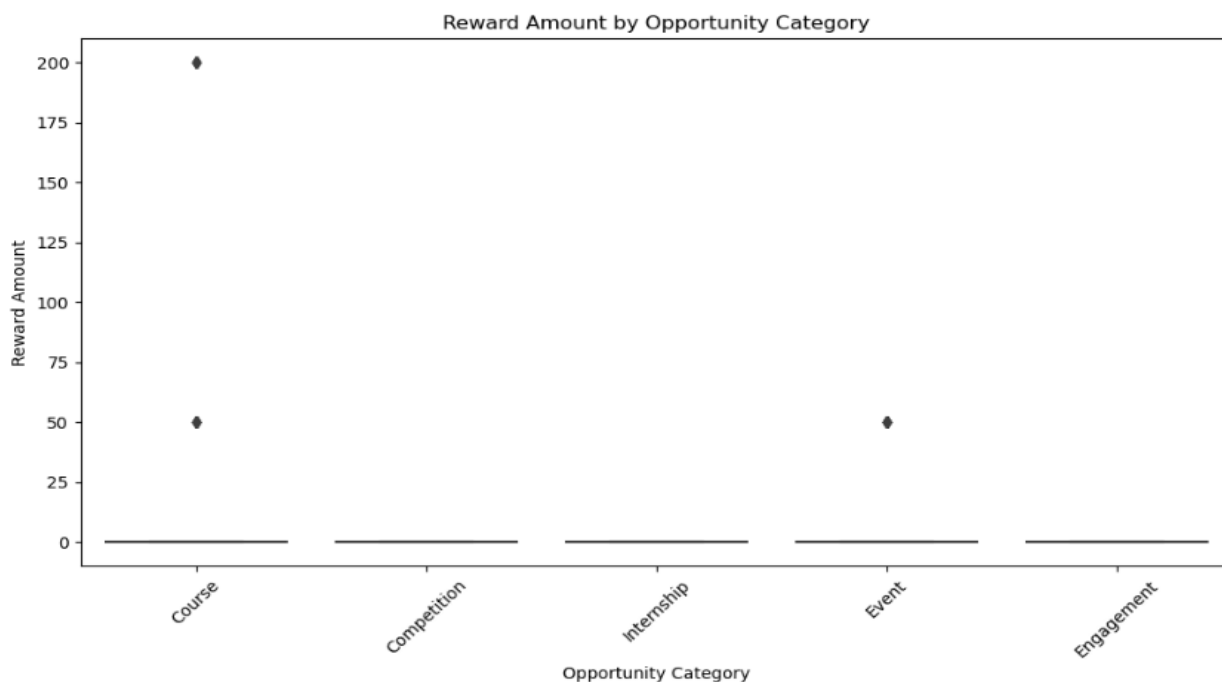
Current Student Status by Current/Intended Major:

- Here these are the various fields that most of the participants are currently pursuing or interested to pursue.
- We can see that in fields like Computer Engineering and Computer Science and Engineering, the Undergraduate Students participation is more when compared to Graduate or high school students.



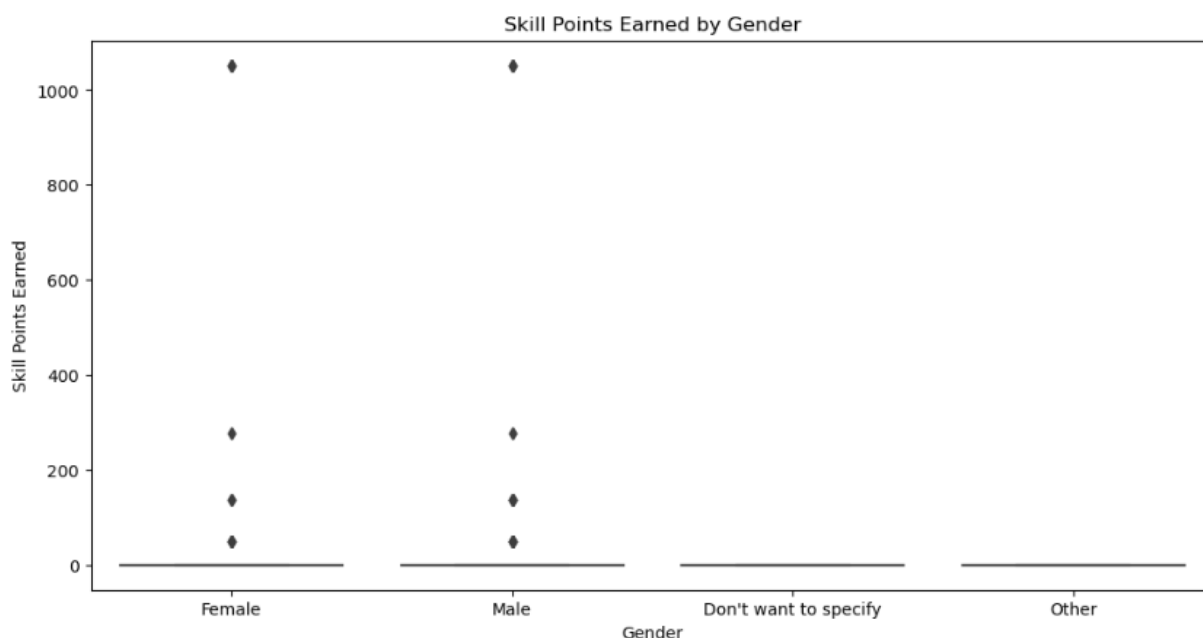
Reward Amount by Opportunity Category:

- Here we can see that until now the opportunity completed students who got rewards are from either Course or Event category and the Reward amount they earned are discrete too i.e either \$50 or \$200



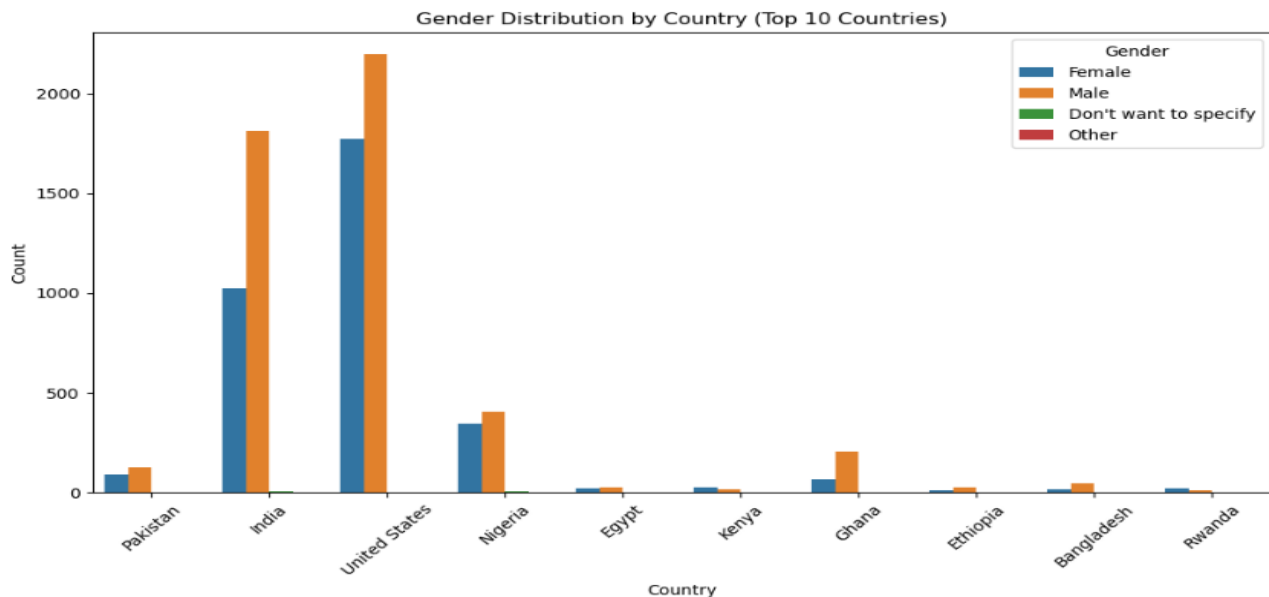
Skill Pints by Gender:

- Here we can clearly see that there is no discrimination in terms of distribution of Skill points based on Gender.



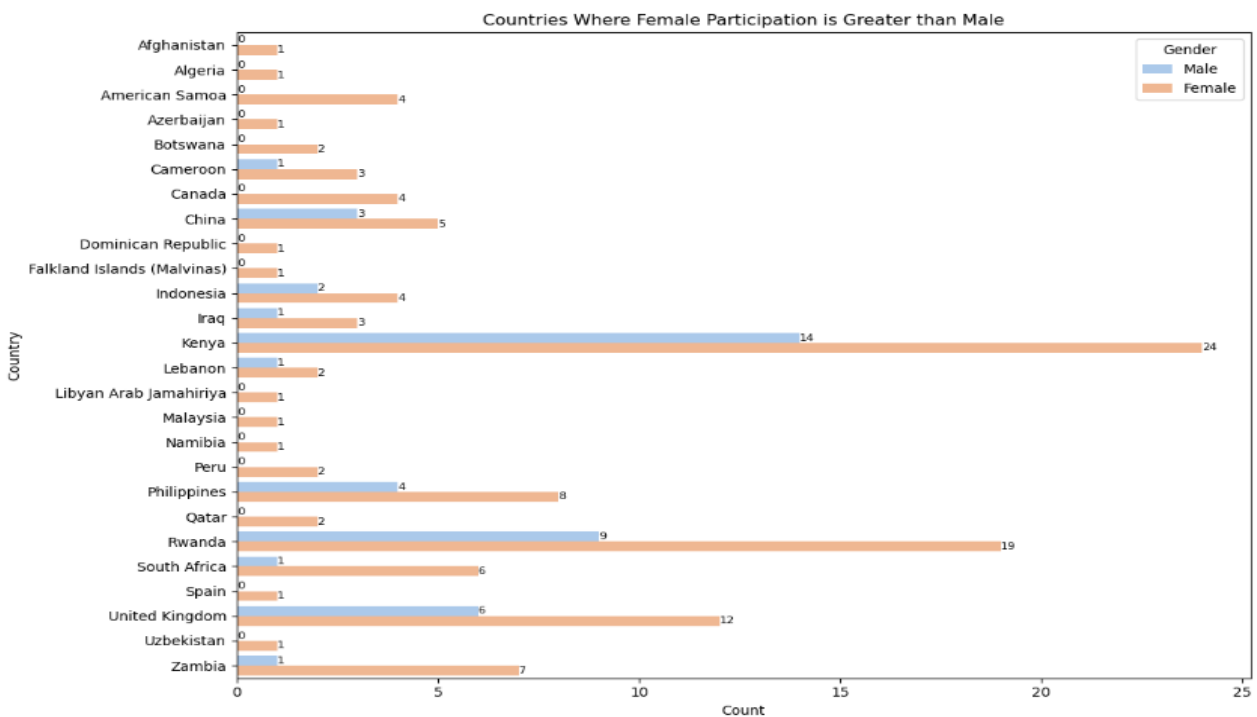
Gender Distribution by top 10 countries:

- Here we can clearly see that there are more number of participants from United States and India, compared to all other countries.
- In Top 10 country Distribution except for Kenya and Rwanda the majority of participants are Male.



Countries where Female participation is greater than Male:

- These are the countries where the participation of females is greater than the males.



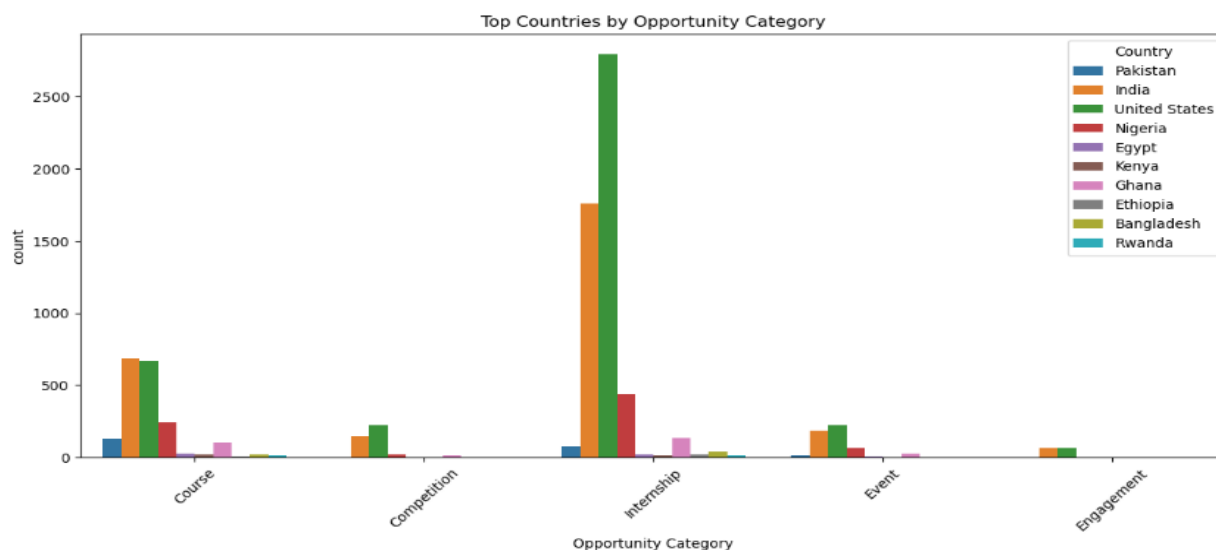
Age vs Reward Amount:

- Here we don't see any Correlation between these two columns, but the people who got awarded are mostly in between 22 to 26 years of age.

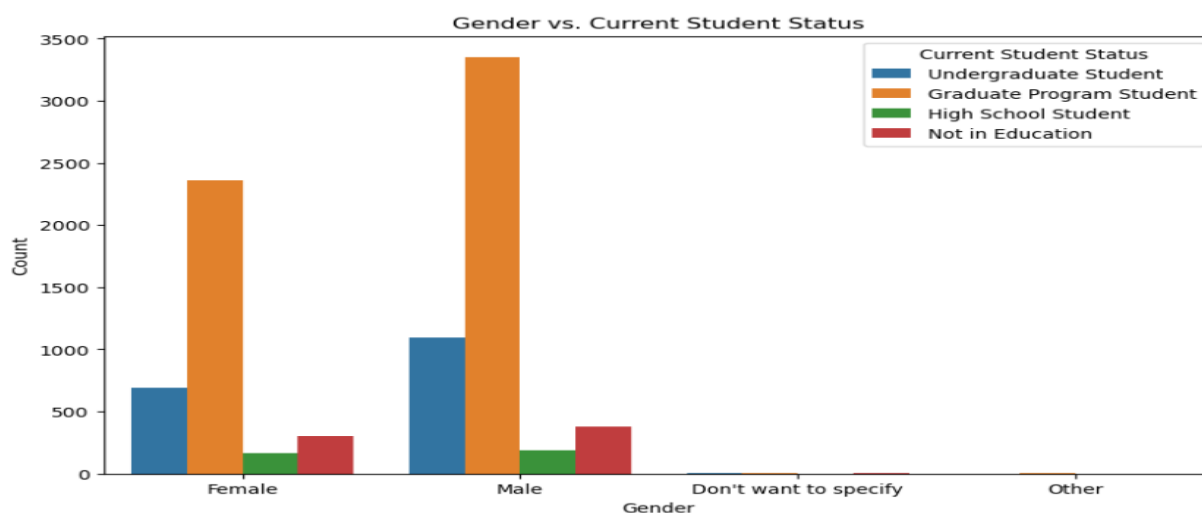


Top Countries by Opportunity Category:

- Apart from Course Category participation from United States is more in all other categories.

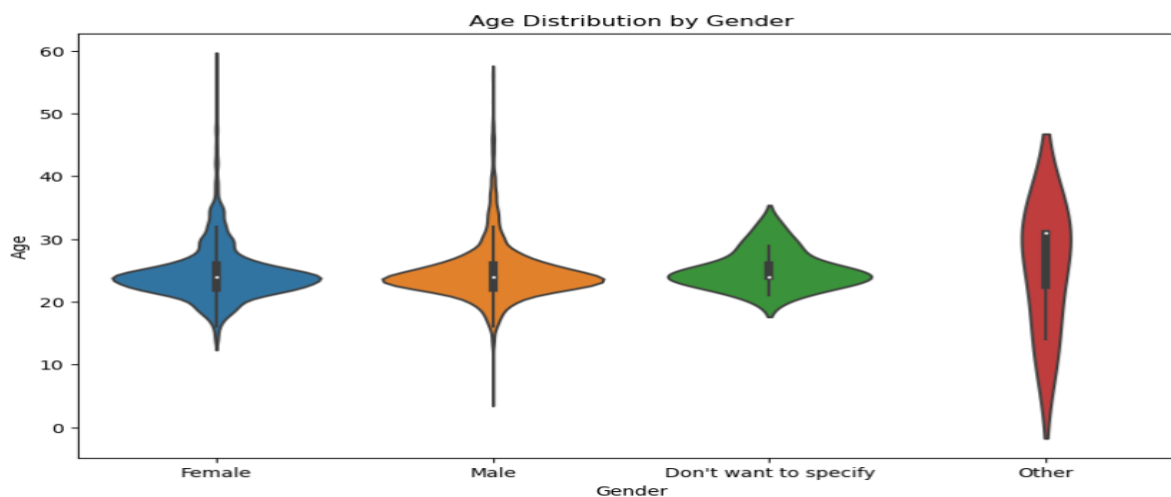


Gender vs Current Student Status:



Age Distribution by Gender:

- Here we can see that most of the distribution across all the genders is in between the 22-26 years, but there are outliers at upper end for both the Male and Females.



Reward Amount vs Opportunity Start Date:

- From the plot we can clearly see that the Reward Amount doesn't depend on when the Opportunity has Started(i.e. we cannot say that The opportunities which were started earliest gives more reward amount or so) . It is based on What opportunity has been completed and on his performance.



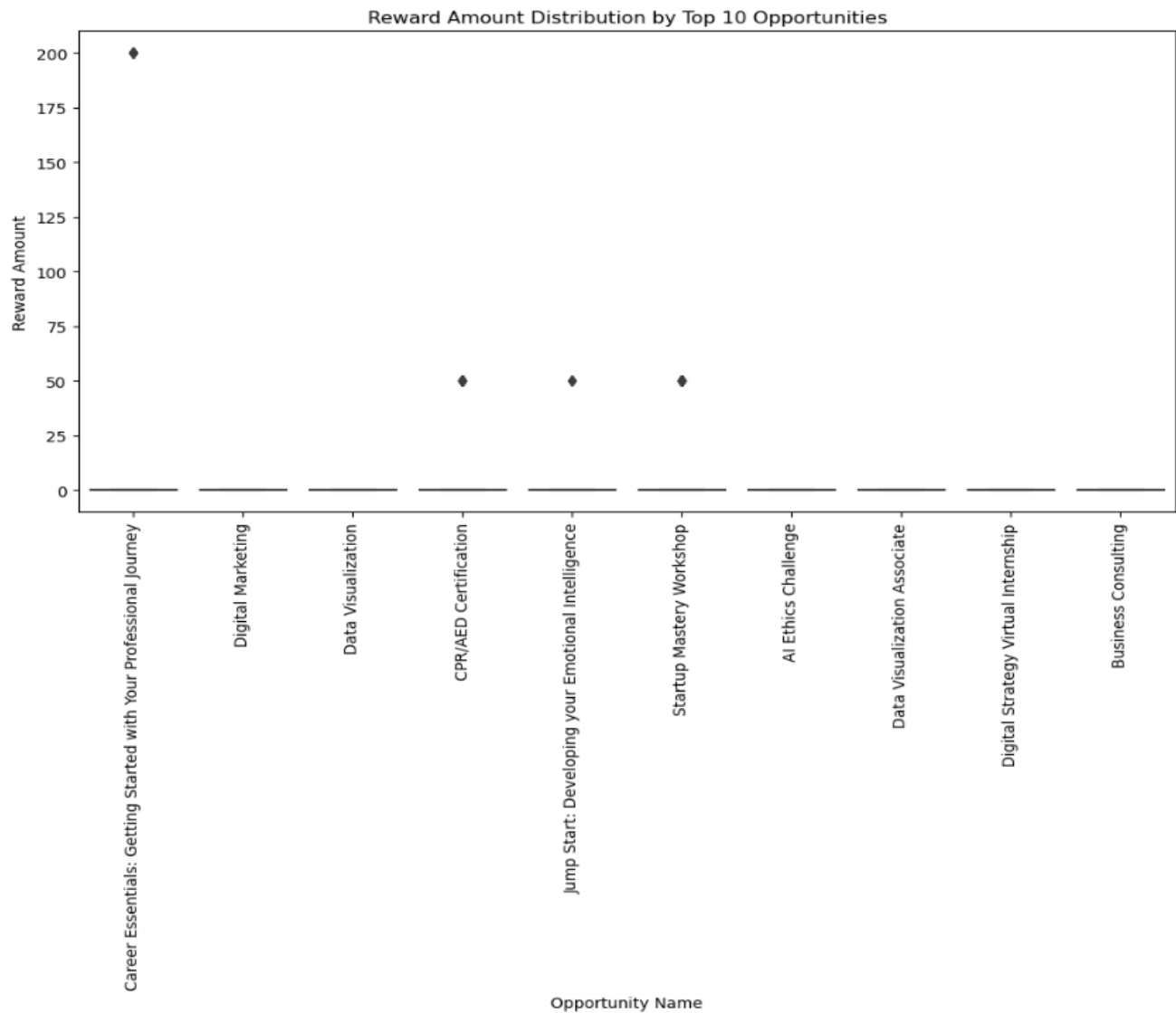
Reward Amount vs Skill Points Earned :

- We can see that there is weak positive correlation between both these columns.



Reward Amount by Top 10 opportunities:

- Here we can see that the distribution of Reward Amount is in the four opportunity names only, indicating that until now those who had completed the opportunities and earned Reward amount, were from these opportunities only

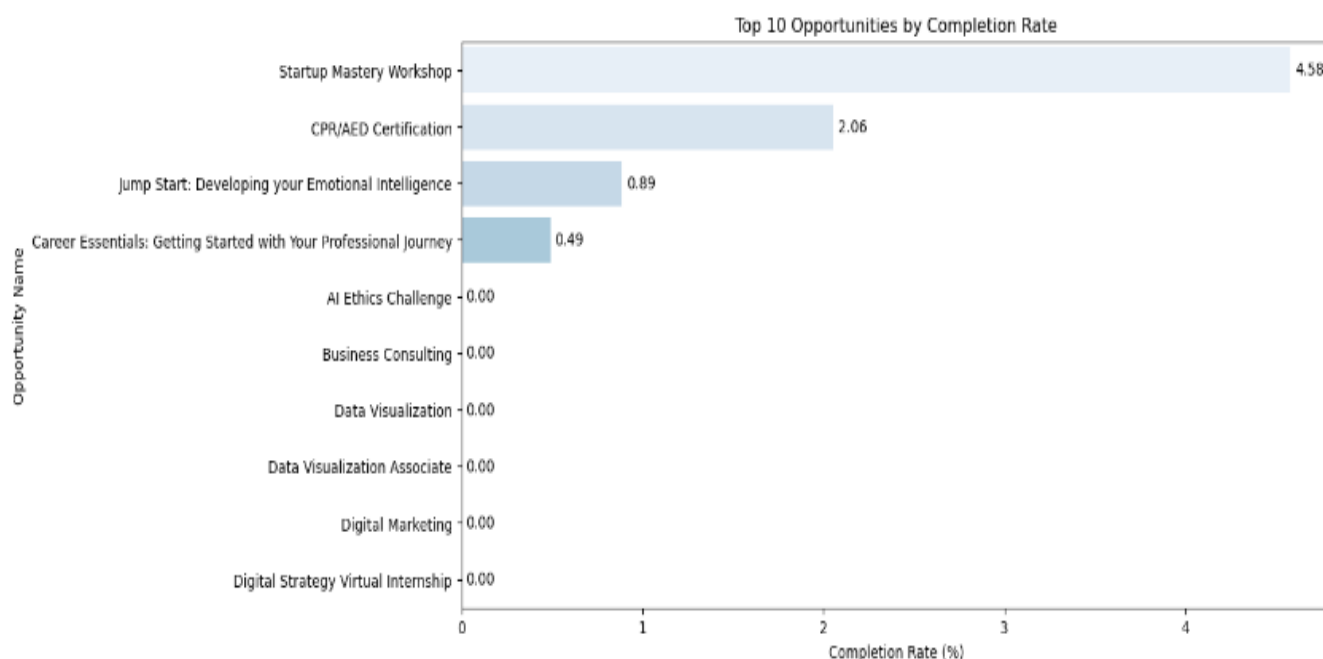


Completion Rate:

- Overall if we see, there is only 29 records out of 8558 records which can be considered Completed, so the overall completion rate is 0.34% only.

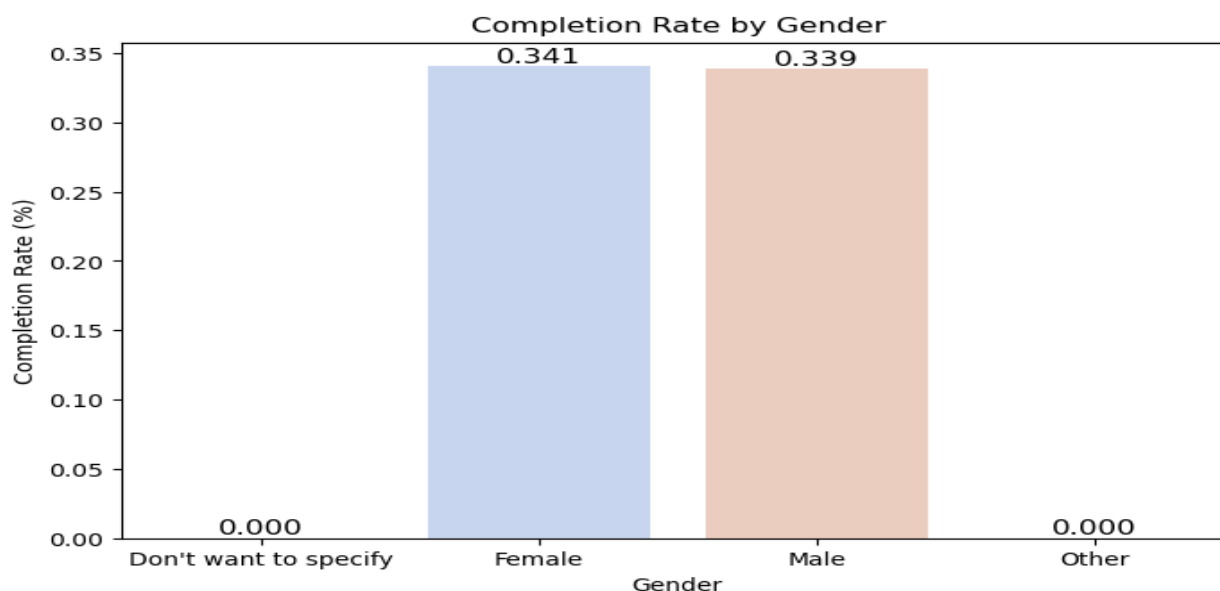
Opportunities by Completion Rate:

- The completion rates across all the opportunities are very low, only these 4 opportunities have been completed so far by some students, rest of them are yet to be completed.



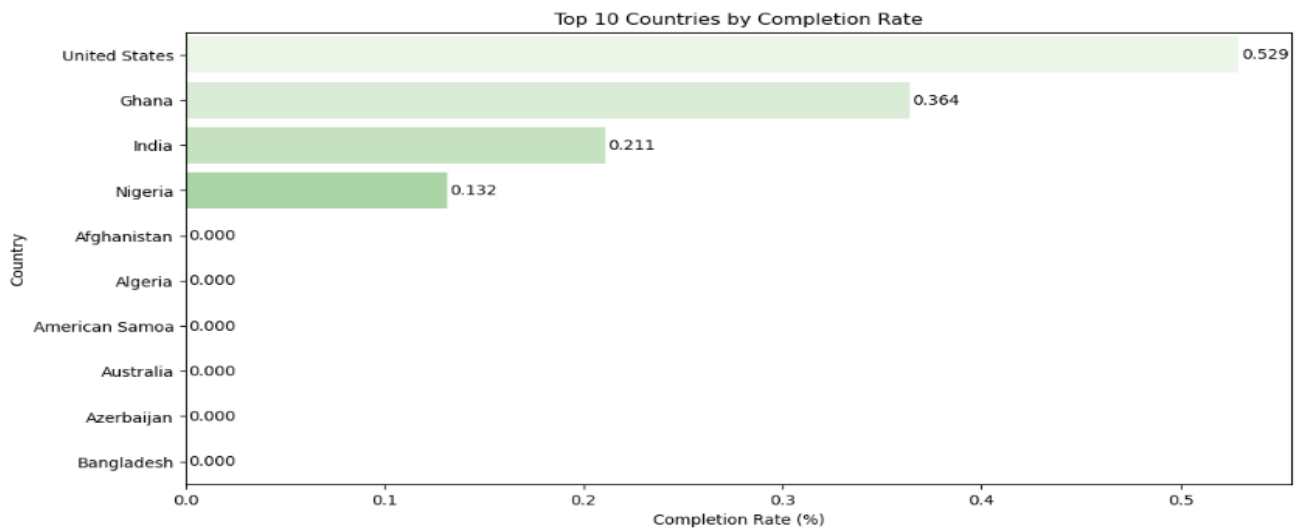
Completion Rate by Gender:

- Although the Completion Rates of Male and Female are almost same, but completion rate of female is slightly higher than male.



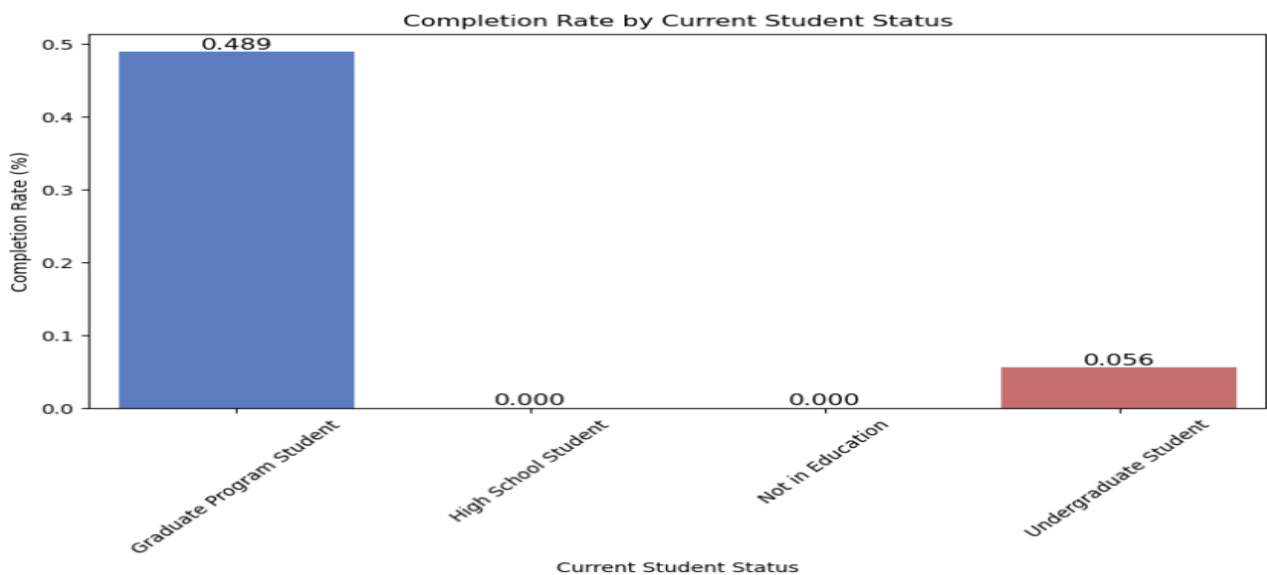
Top Countries by Completion Rates:

- We can see until now people from United States, India, Ghana, Nigeria are the only ones who have completed the opportunities.



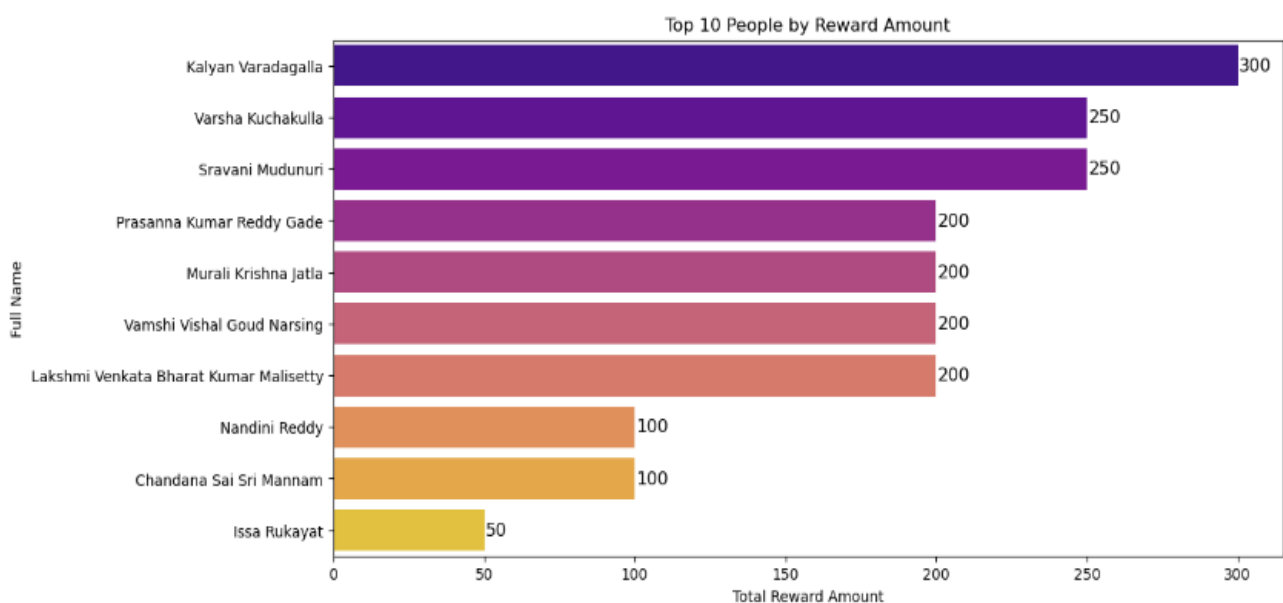
Completion Rate by Current Student Status:

- The completion rates of Graduate Program students is higher compared to all other Student Status.



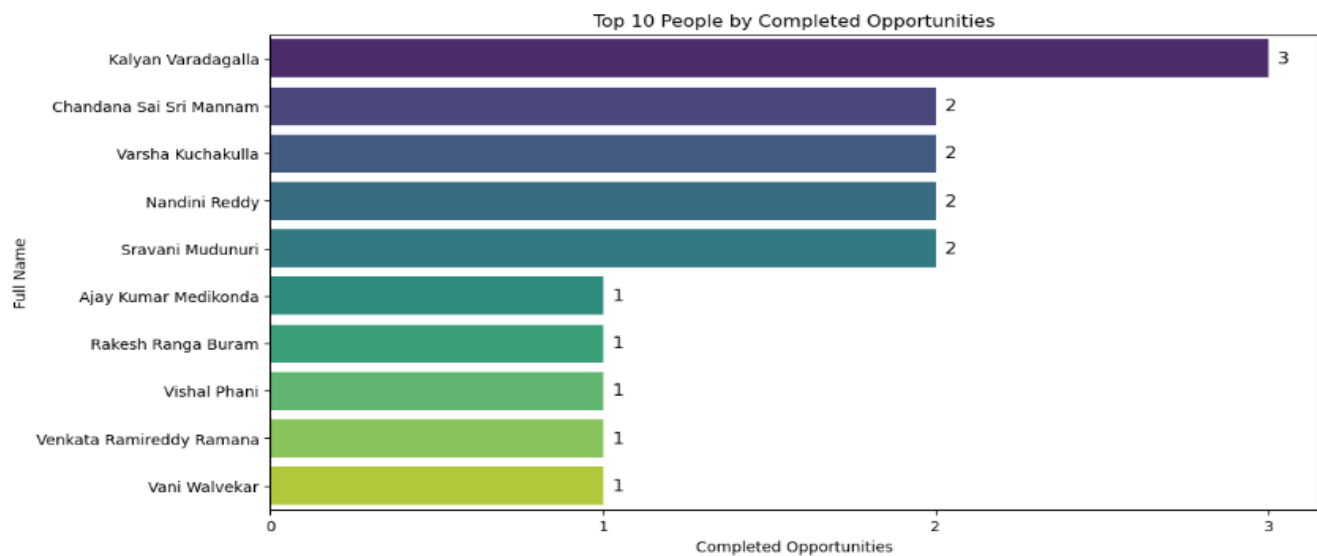
Top 10 People by Reward Amount:

- These are the people who have received overall highest Reward Amount so far (but this may be through more than one opportunity as well)

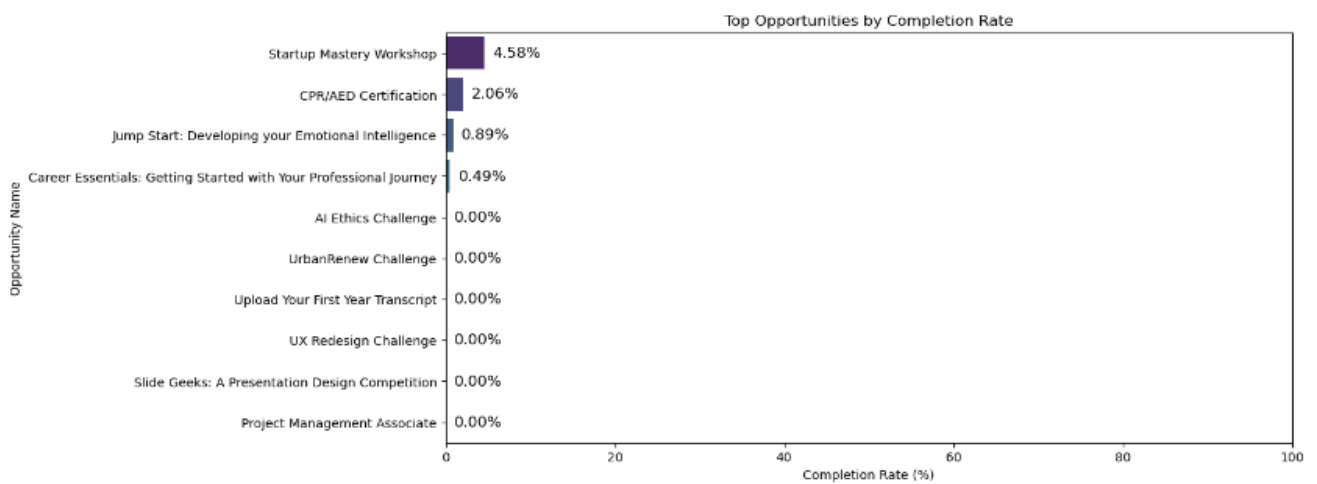


Top 10 people by Completed Opportunities:

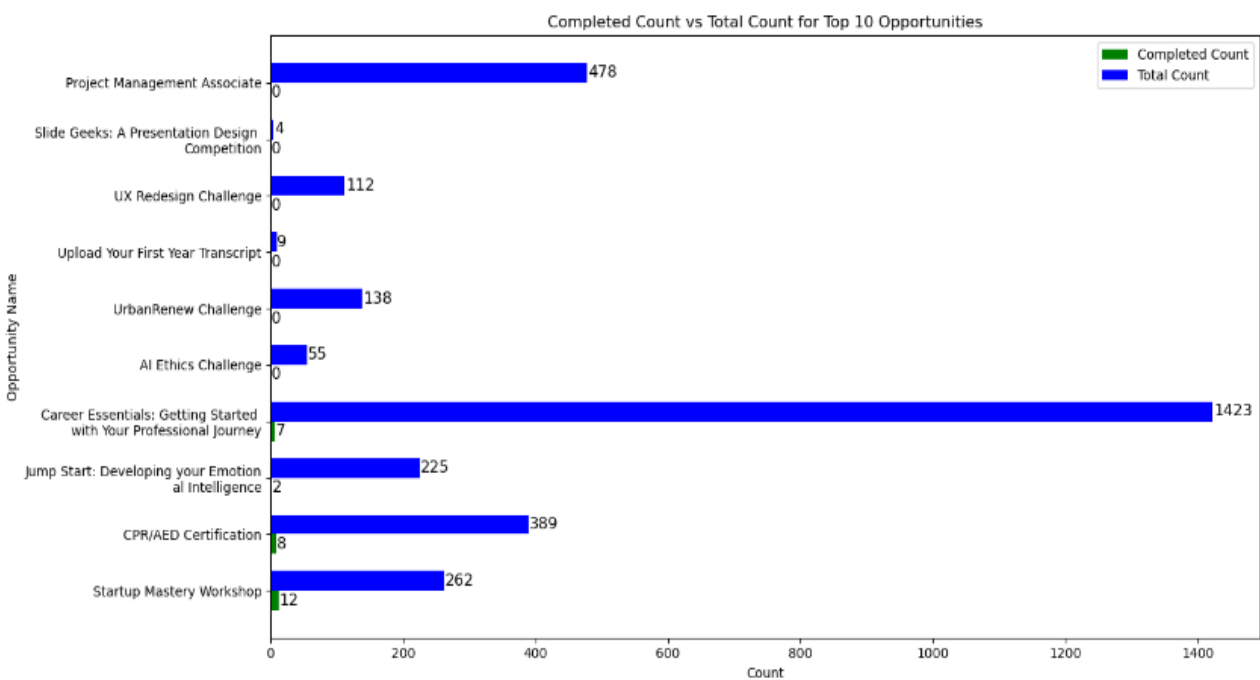
- These are the top 10 people who have completed more opportunities.



Top Opportunities by Completion rates:

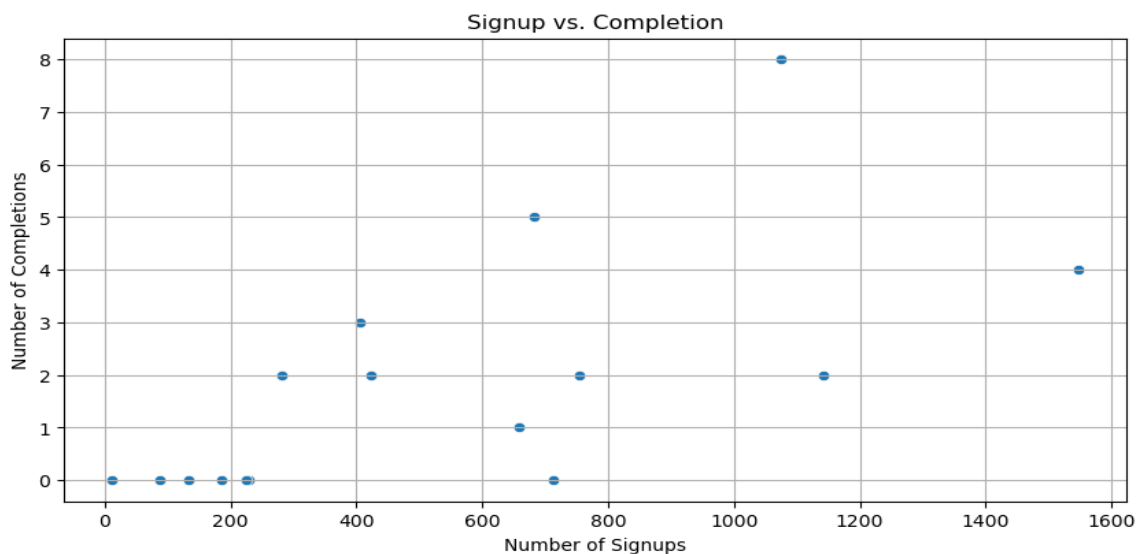


Completed Count vs Total count for top 10 Opportunities:



Signups vs Completion:

- There is a moderate Correlation between Signups and Completion, indicates that as number of signups increases the completions also increases.



Status Description vs Top 10 Opportunity Name:

- Here we can see that for Opportunities like CPR/AED Certification, Carrer Essentials: Getting Started with Your Professional Journey the Rejections are very less compared to all other opportunities.

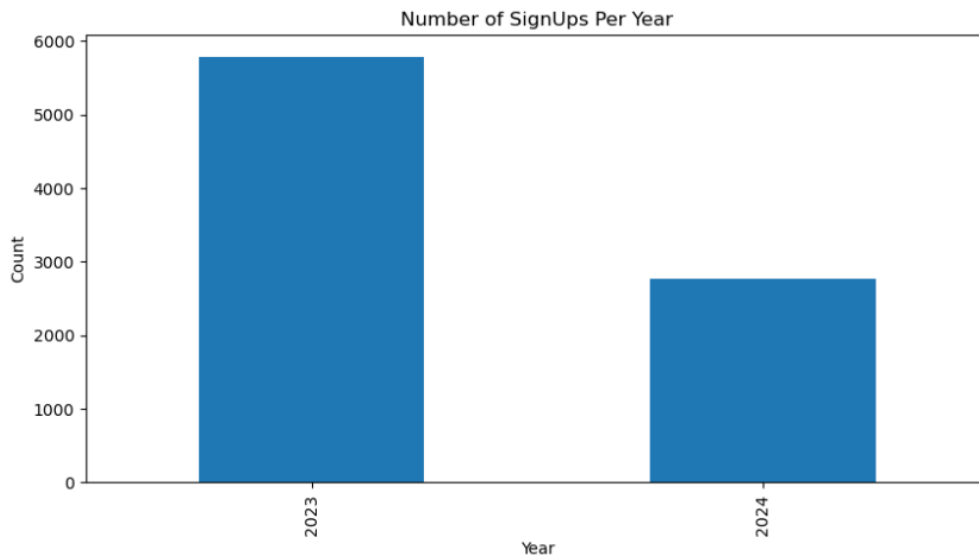


Note: Since there are only 29 have been completed out of 8558 opportunity sign-ups, we might not get many insights or relations from this data, but since there are date-time columns we can find out trends or patterns in those columns.

Trend Analysis:

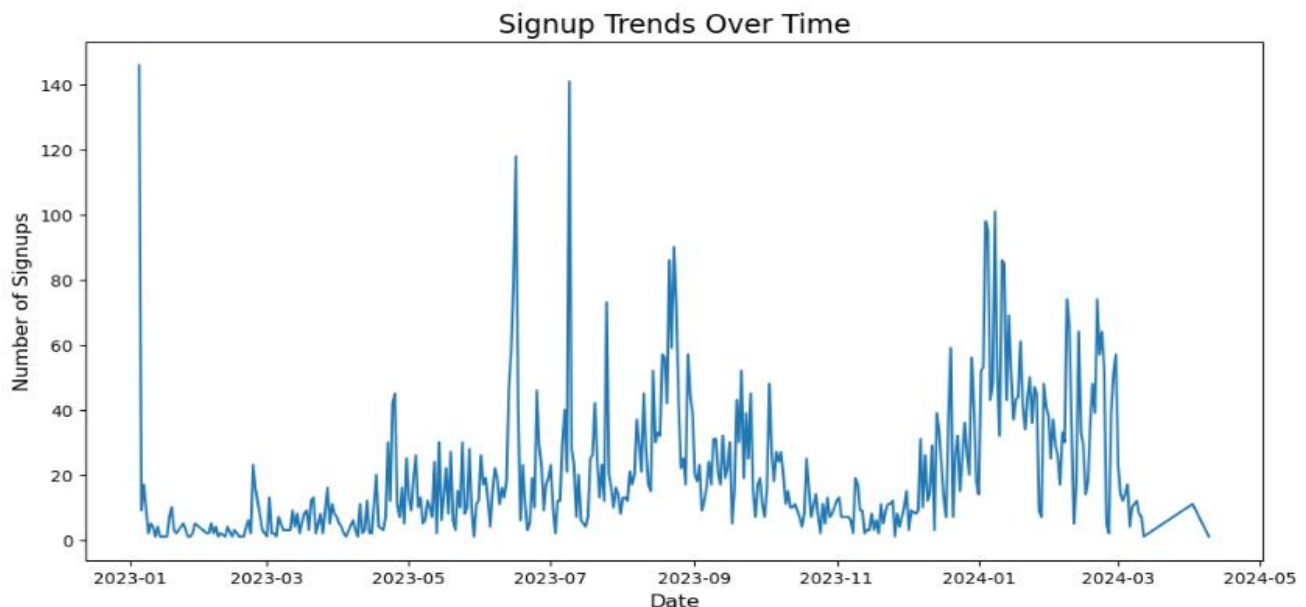
Number of Sign-ups per Year:

- Here, the data is of range from January 2023 to march 2024.
- Even though the Entire Signups in 2023 are approximately around 5700, but I just 3 months of 2024
- The signups are around 2800, indicating that there is an Upward trend in Signups over the years.

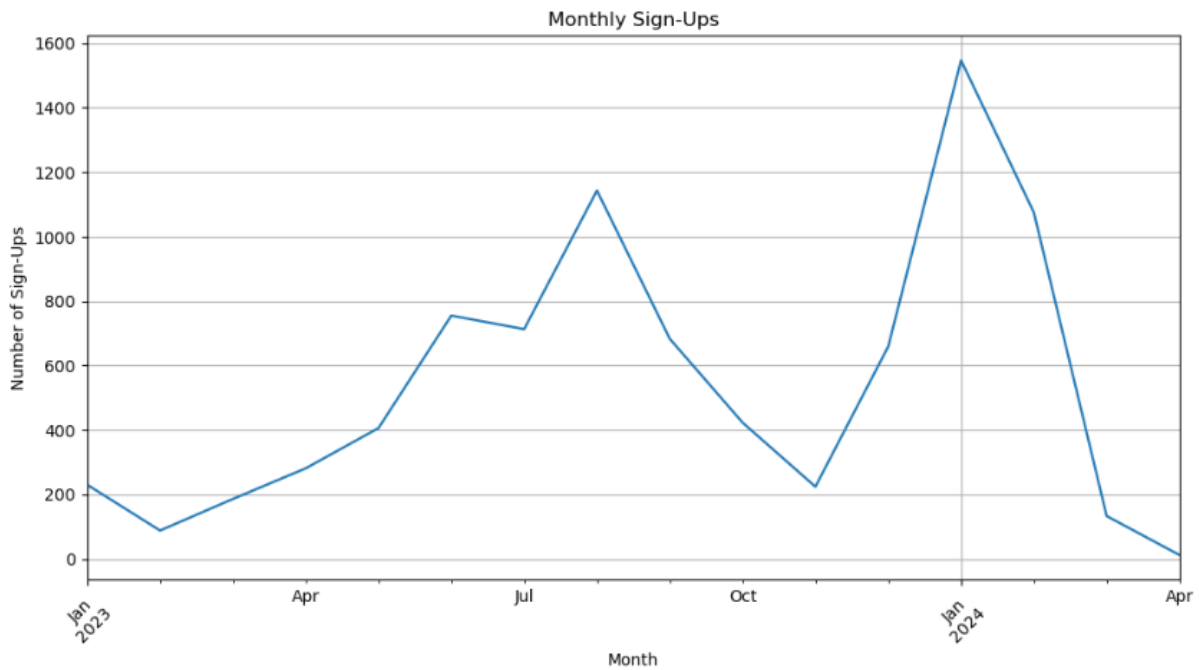


Signup Trends over Time:

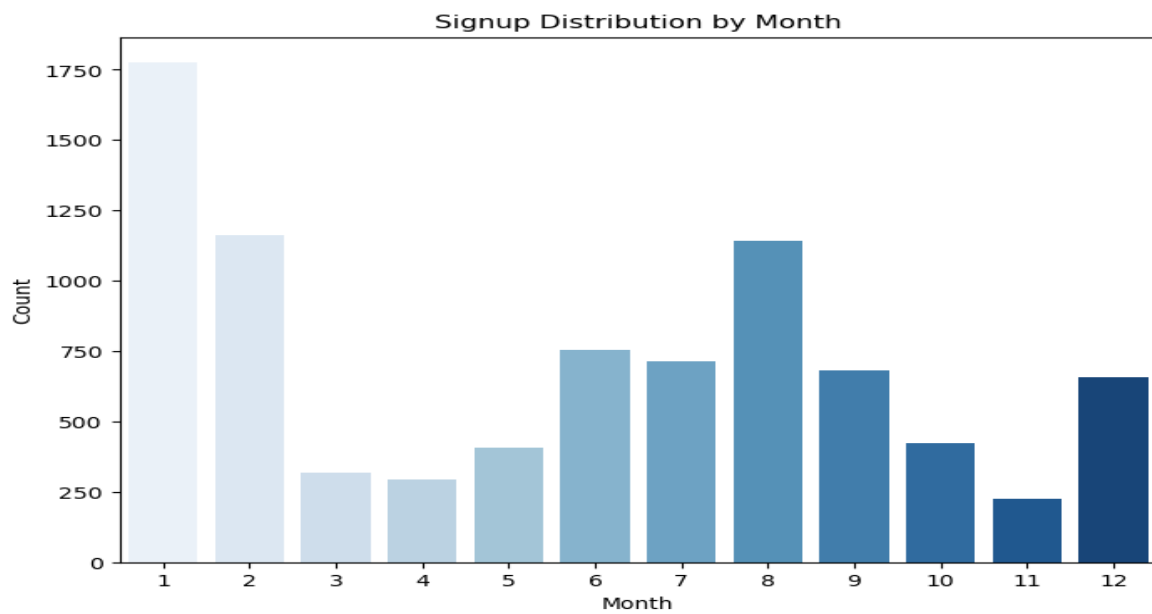
- Although there are fluctuations in signups over the time, the cumulative signups are in upward trend only.
- We can see that there is a cyclic trend, like in months of January, June, July, August months there are huge spikes, indicating high number of sign ups.



- Since there is Huge participation from United States and India only, In January month actually we might assume that that is due to Winter vacations in United States, and In months of June, July, August months that is generally either the Summer vacations or start of the Academic year for Graduate and Under Graduate Students.

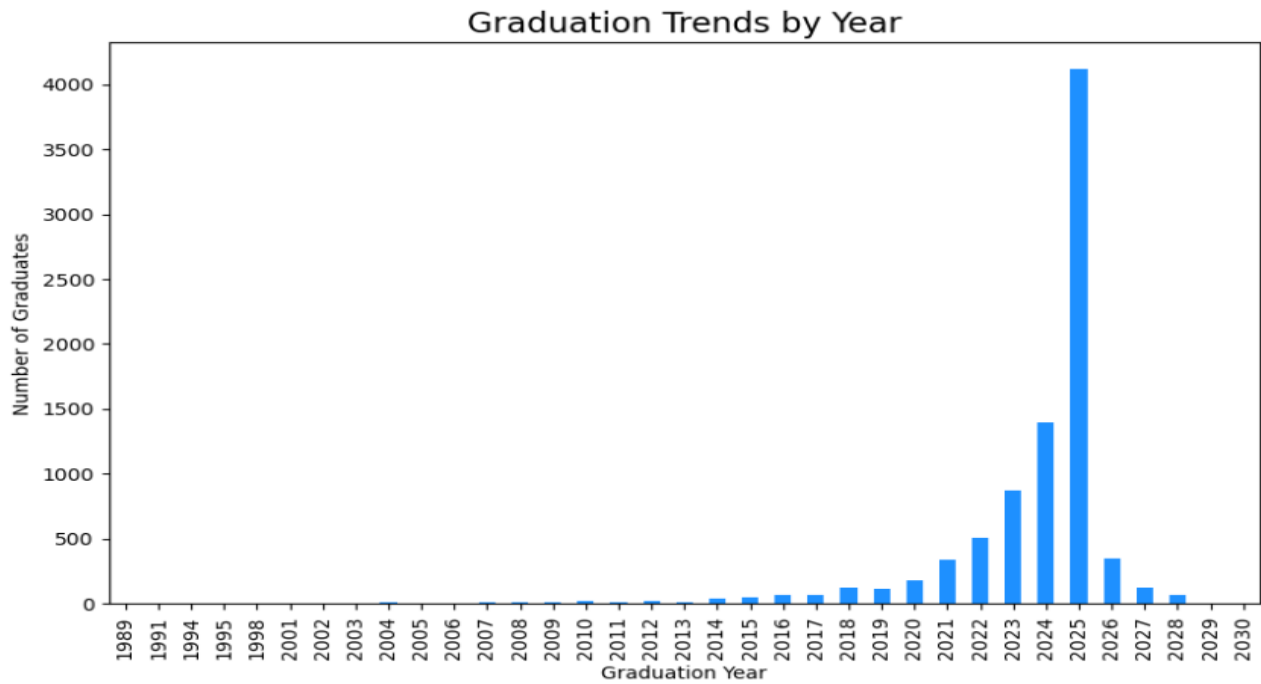


- Here we can clearly see huge spikes in August 2023 and January 2024 months, indicating very high amount of signups in those months.



Graduation Trend by Year:

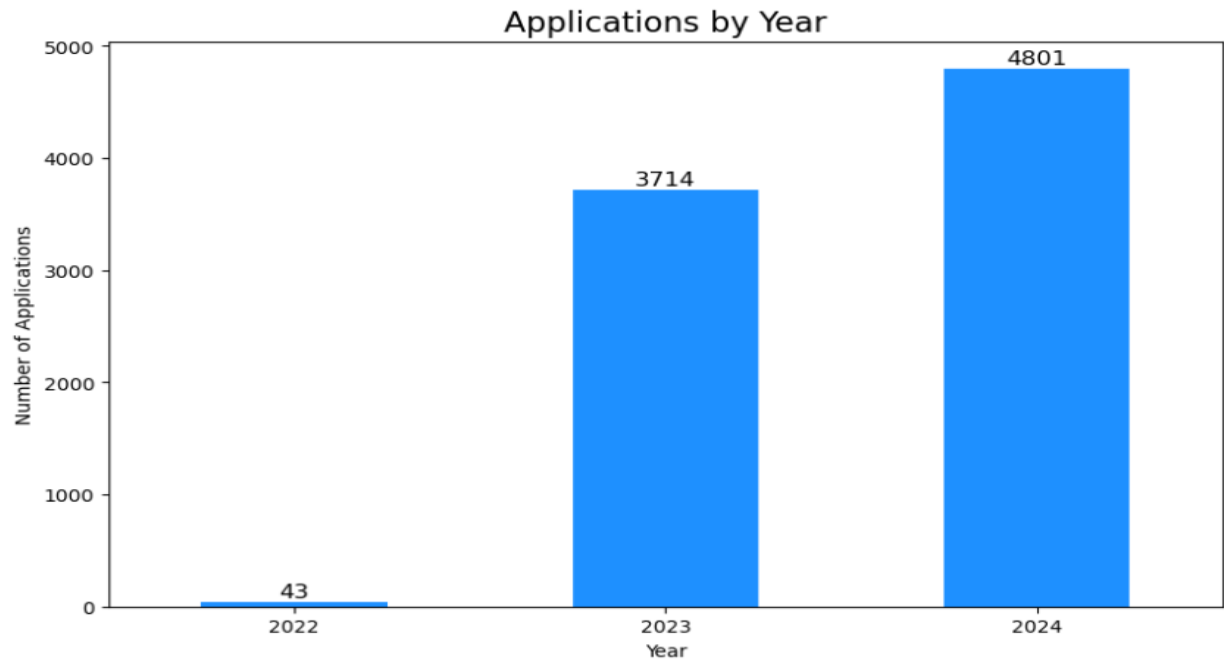
- Here we can see that the cure is Left skewed, indicating that there are some outliers on lower end of the Graduation Year.
- Most of the participants are Graduation Students who have not yet completed their Graduation yet.
- Compared to Graduated there are more graduating students, who are expected to be graduated in next 1-4 years.



Apply Date Trend:

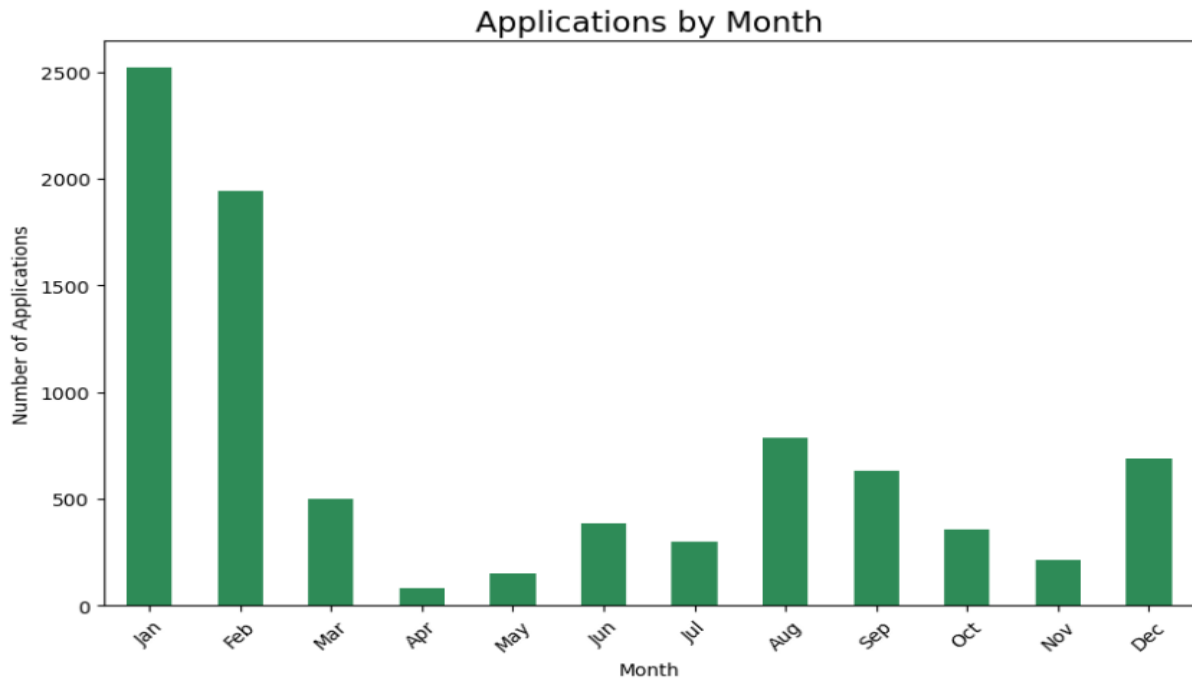
Applications by Year:

- We can see that over the years the number of applications have been increasing enormously, also the 2024 has more Applications, but this 4801 applications in this year are from the first 3 months of the year only, actually we can estimate applications total of 3 times of this value by the end of this year.



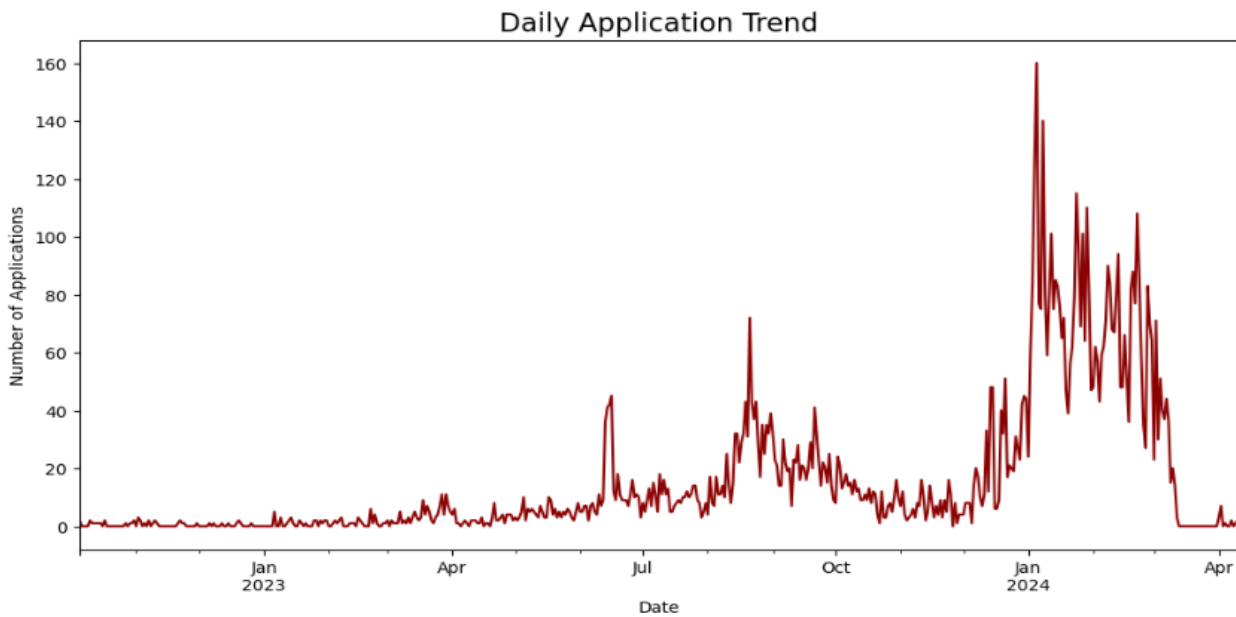
Applications by Month:

- Even here we can see that there are huge number of applications in January, February months, and some small spikes again in August, September and December months.



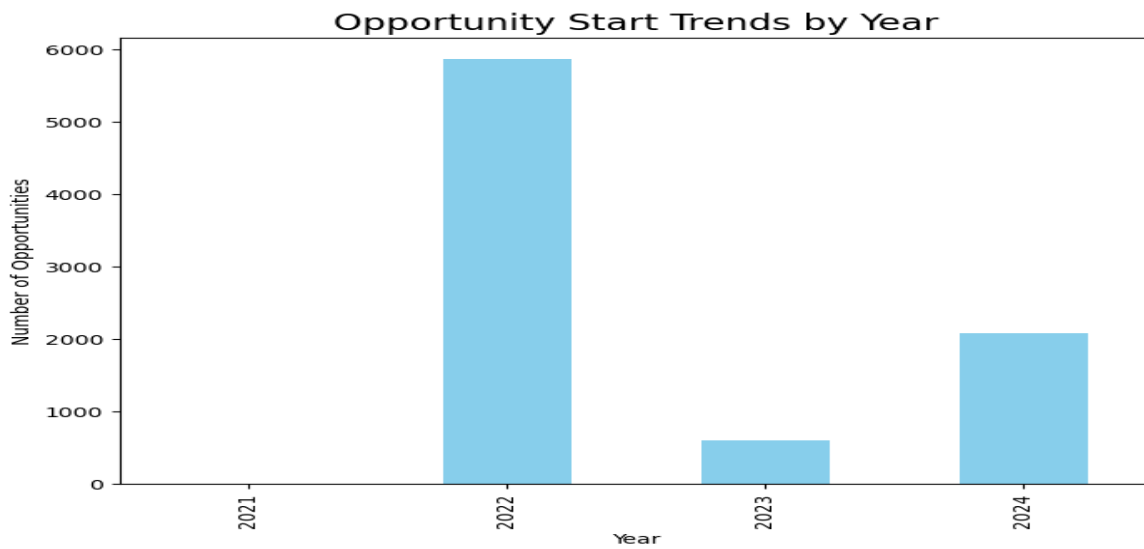
Application Date Trend:

- We can see that the cumulative applications have been increasing over the time and are expected to increase even further in the upcoming time.



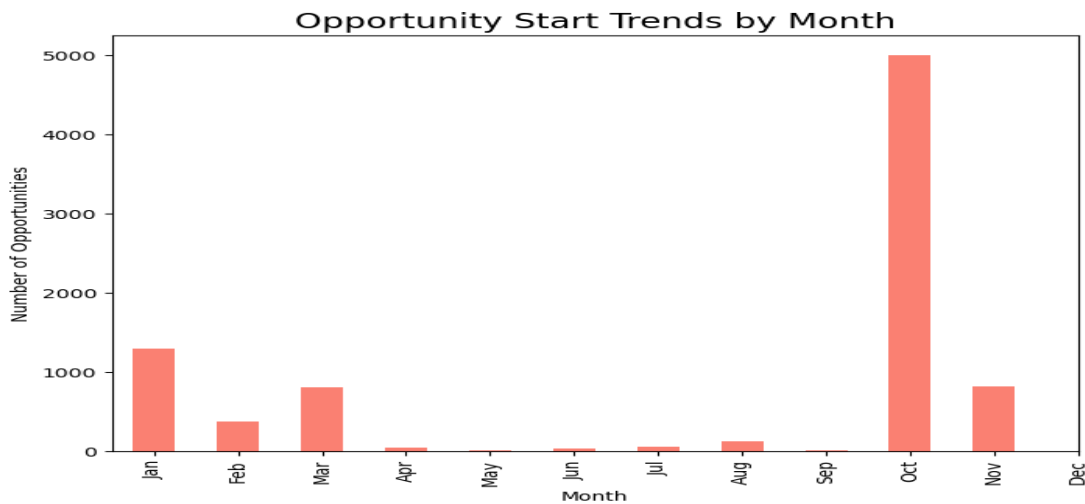
Opportunity Start Date Trend by Year:

- Here we can see that in 2022 year there were more than 5500 opportunities started in this year, but there is sudden downfall in number of opportunities starting in 2023.
- But in 2024 we can again see that the number of opportunities starting are more (In 2023 in just 3 months approximately 2000 opportunities have been started)



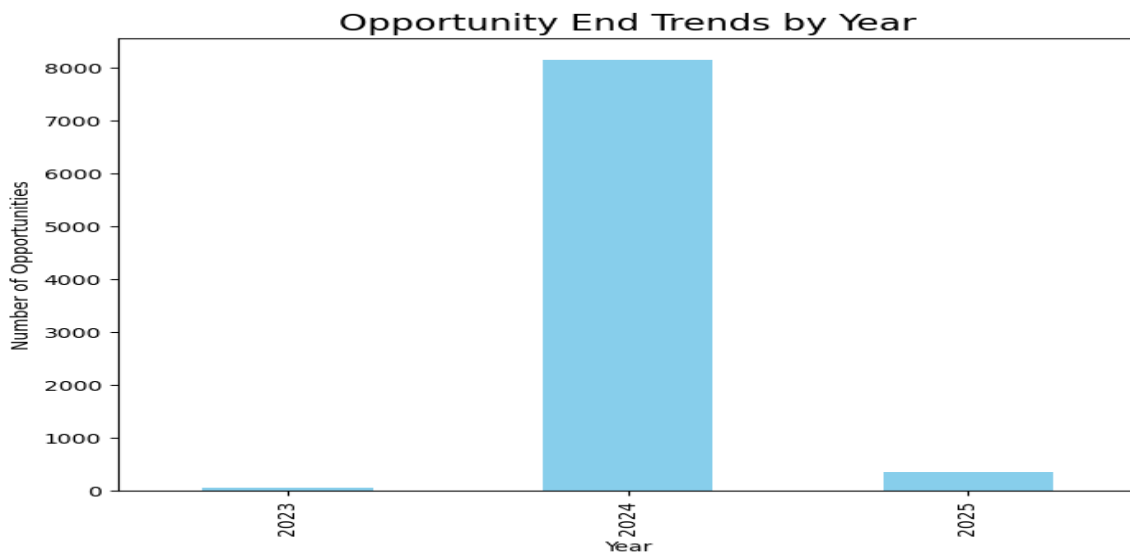
Opportunity Start Trends by Month:

- In the month of October there is huge spike, indicating more number of opportunities starting in October month.
- Also there are few opportunities being started in January, March, and November months



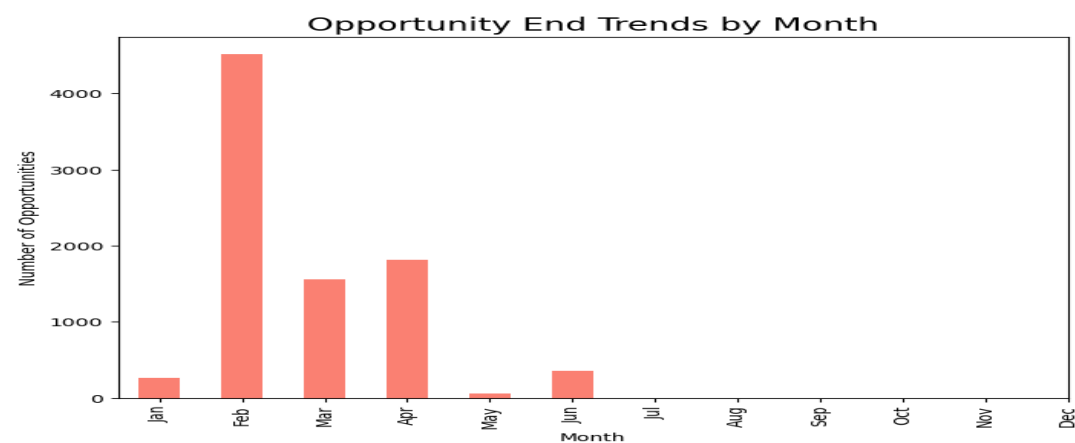
Opportunity End Trends by Year:

- Here we can see that most of the opportunities are being ended in this year i.e. 2024.



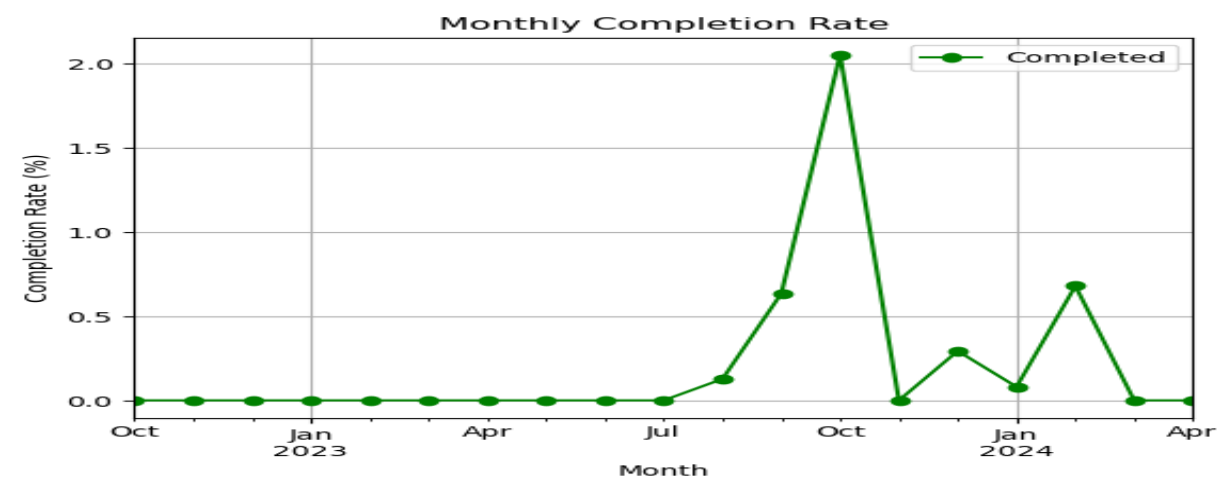
Opportunity End Trends by Month:

- Here we can see that most of the opportunities are being ended in the February, March and April months.



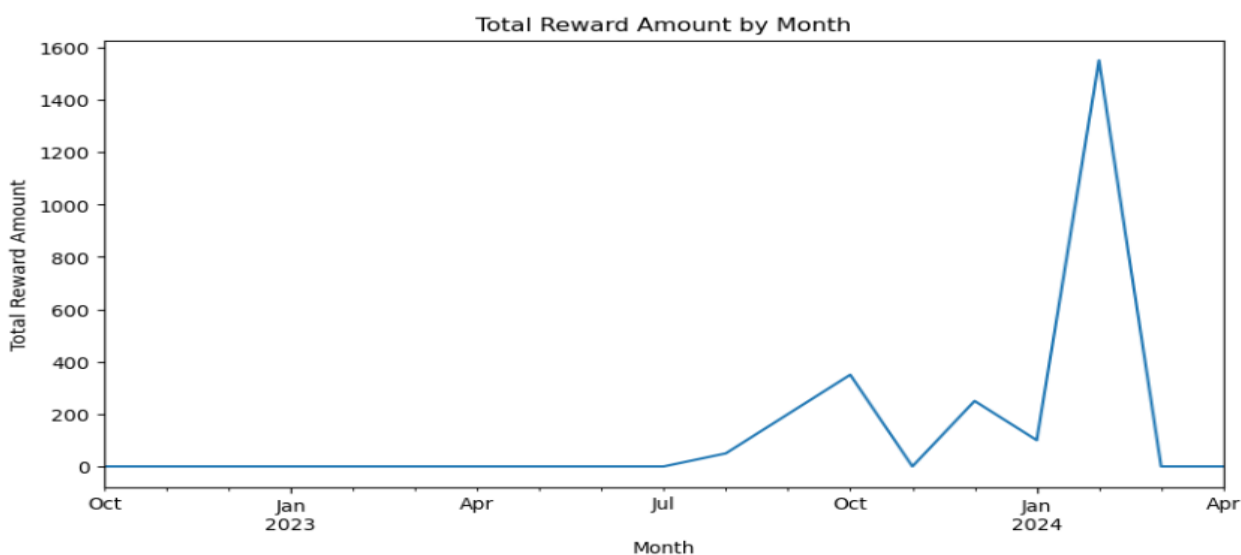
Monthly Completion Rate :

- Here we can see that most of the opportunities are being completed in the months of October, then some of them have been completing in September and February months



Reward Amount by Month:

- Here we can see that most of the completed students were mostly awarded in February month and some of them were awarded in October, September and December months respectively.



Low Completion Months

- Further analysis needs to be done by the company like why there are less opportunities starting in between April to September months and December month too.

Recommendations

Target Peak Months

- Implement marketing strategies targeting peak signup months, such as promotional campaigns or discounts during high-traffic periods.

Further Investigate Drops:

- Actually there is drop in signups in the months of February, March and November months , this might be because of their Examinations in the college.

Segment Strategies:

- Actually apart from Graduate Students and some Undergraduate student others have not yet completed atleast one opportunity, this might be due to various reasons, one of them might be complexity , so we can launch some easy opportunities, which can be easier for High school students and Not in Education Students.

Conclusion

Key Insights

The analysis revealed that:

- User signups peak during June to September, January to February months.
- There is an Upward trend in Signups over the years.
- Applications have surged significantly in early 2024, surpassing the total for 2023, indicating growing interest and the need to scale efforts efficiently.
- Most Application Dates are also from the January to February months followed by August to September months.
- Opportunities saw a decline in 2023 but rebounded sharply in 2024, with around 2,000 launched in just the first three months.
- In the month of October there is huge spike in Opportunity Start Date, indicating more number of opportunities starting in October month.
- Most of the Opportunity End dates are in 2024, specifically in February to April months.
- When it comes to completed opportunities there is a spike in October month indicating more number of completions in this month, followed by September and February months.
- Reward Distribution has been started in July 2023, it also coincides with increase in Signups too, so the scholarships might be one of the reasons due to increase in number of signups.